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A physics-aware autoregressive encoder–decoder for fast surrogate prediction of transient melt pool dynamics in laser welding and additive manufacturing

Zhehao Zhang ^a, Zihan Wang ^a, Xuefei Cui ^b, Yi Zheng ^c , Daqing Wang ^{d e} [Show more](#) ✓

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1. Introduction

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5. Conclusions

Highlights

- Physics guided network predicts melt pool heat and flow in laser powder bed fusion.
- Process and material parameters are embedded for robust generalization.
- Step by step training enforces temporal consistency in sequence prediction.
- Millisecond predictions support design, monitoring, and digital twin uses.

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Abstract

A formidable challenge in laser welding and additive manufacturing lies in forecasting coupled thermal–fluid dynamics, specifically the Marangoni-driven melt pool behavior. While high-fidelity numerical simulations are computationally prohibitive, purely data-driven surrogates often fail to generalize across different physical conditions. Here, we introduce the Physics-Aware Autoregressive Encoder–Decoder Network (PAED-Net), a framework that integrates material properties and process parameters via a physics-aware subnetwork and leverages an adaptive multi-task autoregressive strategy to predict the transient evolution of pressure, temperature, velocity, and volume fraction fields within the micro-melt pool. Validated against a Marangoni benchmark and complex multiphysics simulations, the model maintains a relative error of below 8% across diverse materials and process conditions. Notably, PAED-Net successfully resolves the dynamics of micro-scale Marangoni vortices, flow structures induced by temperature gradients, and the associated coupled heat and mass transfer. Compared to conventional finite-element baselines, PAED-Net achieves millisecond-level inference per pass with speedups of several orders of magnitude. This unified, parameter-conditioned framework enables rapid scenario evaluation and process optimization, delivering robust generalization for complex Laser Welding and Additive Manufacturing processes.

Keywords

Melt pool thermofluid modeling; Physics-aware surrogate modeling; Autoregressive neural networks; Parameter conditioning; Marangoni convection

CRediT authorship contribution statement

Declaration of competing interest

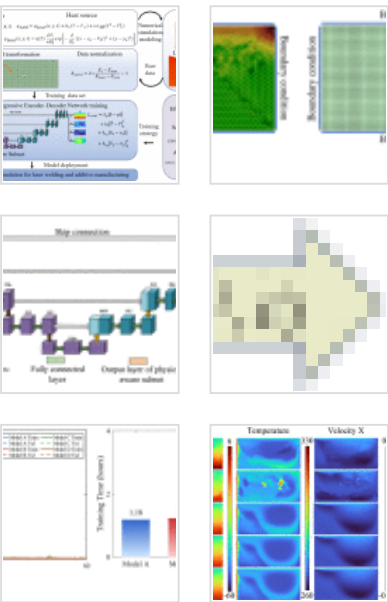
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Metal additive manufacturing (AM) has been widely adopted in aerospace, energy equipment, and biomedical sectors, serving as a key route for rapid fabrication of complex components [1], [2]. Relying on high energy density laser heat sources to achieve localized melting and solidification, it offers high geometric freedom and material utilization [3]. However, the process exhibits strongly nonlinear and highly transient coupling among heat transfer, fluid flow, and phase change [4], [5]. Under laser or arc input, steep temperature gradients and recirculating flow develop in the melt pool. Among the underlying thermo–fluid mechanisms, Marangoni convection is a key factor governing melt–pool morphology, flow behavior, and heat/mass transport. Through its interplay with thermophysical properties and process parameters, this effect exerts a major influence on process defects such as porosity, lack of fusion, and cracking, as well as on grain microstructure [6], [7], [8]. Therefore, systematically investigating the coupling among the Marangoni effect, heat transfer, fluid flow, and phase change at the melt–pool scale is a prerequisite for defect suppression and precise control of the microstructure–property relationship in AM [9], [10]. The combination of in situ experimental characterization and high-fidelity multiphysics simulation provides an effective pathway to deepen the mechanistic understanding of metal AM [11]. Although multi-source in situ monitoring such as visible/infrared imaging, synchrotron X-ray, temperature sensors, and acoustic signals captures surface morphology and partial thermal information [12], [13], comprehensive and continuous measurement of internal multiphysics fields in the melt pool remains difficult, requiring high-fidelity numerical models for supplementation.

Finite element method (FEM) and finite volume method (FVM) based numerical models for AM and welding can resolve the evolution of temperature and velocity fields, phase change, and the morphology of the free surface and keyhole, thereby capturing complex phenomena such as Marangoni convection, vapor recoil, powder melting, and wetting [14]. These models have become essential tools for process optimization, defect prediction, and microstructure control [15], [16]. In recent years, high-fidelity, melt–pool–scale multiphysics models for AM have advanced rapidly. Representative multiphase AM [17], Sui et al. [16], and Luo et al. [18] explicitly incorporate

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temperature-dependent properties, Marangoni shear stress, vapor-recoil pressure, free-surface deformation, and keyhole evolution under single-/multi-track and multilayer powder-bed conditions. Wei et al. [19] further proposed a part-scale multiphysics framework that couples heat conduction, melt-pool fluid flow, and residual-stress/microstructure evolution, enabling efficient simulations that balance predictions of melt-pool morphology, defects, and microstructure. In laser welding, Li et al. [20] employed a model with temperature-dependent surface-tension coefficients and Marangoni boundary conditions to study how micro-texturing at metal–polymer interfaces regulates melt-pool flow, interfacial wetting, and filling behavior. From the perspectives of physical processes and heat source modeling, Jiménez-Xamán et al. [21] systematically reviewed conduction mode and fully coupled multiphysics laser welding models, particularly modeling strategies that include Marangoni convection, vapor recoil, keyhole dynamics, and free surface tracking. However, simultaneously accounting for free-surface tracking, phase change, temperature-dependent properties, radiative transfer, and Marangoni effects makes these simulations highly sensitive to mesh resolution and time step size. Even short physical durations often require wall-clock times of hours to days, limiting applicability to engineering scenarios that demand extensive repeated computations, such as parameter optimization, uncertainty quantification, or online monitoring [22]. Moreover, model construction and solution rely on intricate physical assumptions and numerical settings, requiring substantial modeling effort, which constrains industrial adoption [23], [24].

To reduce the modeling burden and computational cost of numerical simulation in metal additive manufacturing, a machine learning based surrogate modeling paradigm has emerged. Among these approaches, physics driven reduced order models (ROMs) are representative: techniques such as proper orthogonal decomposition (POD) projection compress the state space and reduce the degrees of freedom while retaining dominant physics. Building on powder-scale high-fidelity laser welding and additive manufacturing models, Kizhakkinan et al. [22] integrated tensor-train decomposition with Gaussian

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to construct a multi-fidelity surrogate that enables rapid search in

high-dimensional process spaces. Within the multiphysics object oriented simulation environment framework, Yaseen et al. [23] combined operator learning to realize a non-intrusive ROM for fast and accurate prediction of temperature and solid fraction.

Chaudhry et al. [24] compared POD with a convolutional autoencoder for thermomechanical modeling in selective laser melting, while Peng et al. [25] and Ghnatios et al. [26] coupled POD and RBF with digital twins and process-scale distortion analysis to rapidly evaluate thermal fields and residual stresses across the process window. In addition, Ogoke et al. [27] used a low fidelity simulation combined with a diffusion generative model to upsample coarse melt-pool fields toward high-fidelity quality, reducing analysis cost by roughly two orders of magnitude. Overall, ROM and multi-fidelity methods are physically interpretable, but they rely on problem-specific bases and still face challenges in strongly coupled free-surface, phase-change, and fully three-dimensional melt-pool problems.

Physics-informed neural networks (PINNs) and related methods embed governing equations and boundary conditions into neural networks [28]. By writing the residuals of mass, momentum, and energy conservation into the loss and using collocation sampling with forward inference, they replace the mesh generation and iterative solves required by traditional FEM/FVM. Once trained, a single prediction is typically completed within milliseconds to seconds. Zhu et al. [29] incorporated the mass, momentum, and energy equations into a PINN for a three-dimensional metal AM benchmark and handled boundary conditions with a Heaviside function, enabling a single network to jointly predict temperature and flow fields. At the thermal scale, Xie et al. [30] and Li et al. [31] built PINN models from the heat equation and, using only a small number of FEM samples, achieved high-accuracy prediction of the three-dimensional temperature field and melt-pool dimensions in DED and laser cladding. For online monitoring and control, Sajadi et al. [32], [33] combined ConvLSTM, online learning, and in situ pyrometry to reconstruct two-dimensional temperature fields in real-time or near real-time. To address training cost and convergence issues, Chiu et al. [34] proposed CAN-PINN, which couples

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magnitude speedup in convergence on several fluid-dynamics cases. Overall, PINN strikes a compromise between physical consistency and modeling flexibility, but performance is highly sensitive to network architecture and loss weighting; training remains expensive with unstable convergence, and extension to complex multiphysics melt-pool problems is still limited. Recent studies also note that PINN-type approaches are most suitable for systems governed by well-defined and fixed PDE formulations, while complex manufacturing processes involving evolving material states and strongly coupled multiphysics interactions may not always be easily represented within a fixed PDE-constrained learning framework. Therefore, exploring physics-aware data-driven surrogate models provides an alternative pathway for efficient prediction of such complex processes. While PINN and ROM-based methods have demonstrated strong capabilities in many thermo-fluid problems, their application to strongly coupled melt-pool dynamics with moving interfaces and complex multiphysics interactions remains challenging. These challenges motivate the exploration of alternative surrogate strategies in this work.

Fully data-driven surrogate models learn mappings from process parameters, scan paths, and local history fields to full-field quantities using neural network architectures such as convolutional encoder–decoder frameworks. Compared with traditional numerical solvers, these approaches avoid mesh generation and iterative PDE solving, enabling much faster inference while capturing the dominant nonlinear relationships in multiphysics manufacturing processes [35]. Pham et al. [36] and Le et al. [37] demonstrated that neural-network surrogates trained on high-fidelity finite element results can efficiently approximate thermal responses in additive manufacturing. Their main contribution was to show that machine learning can retain reasonable predictive accuracy while dramatically reducing the cost of repeated simulations, thereby making large-scale process exploration and optimization more practical. Huang et al. [38] and Ogoke et al. [39] used convolutional neural networks to map process parameters and surface infrared temperature fields to three-dimensional melt-pool and keyhole

Typesetting math: 100% g near real-time visualization of melt depth and keyhole morphology

during laser welding and additive manufacturing. Mozaffar et al. [40], as well as Roy and Wodo [41], employed recurrent or feedforward networks to map energy input trajectories and local temperature history to high-dimensional thermal histories for rapid temperature prediction and quality assessment. These methods offer notable advantages in inference efficiency and modeling simplicity; however, lacking explicit physical constraints, they depend strongly on the training data distribution and show limited generalization across materials, processes, and out-of-distribution operating conditions.

Taken together, recent advances show that ROM, PINN, and purely data driven surrogate models can substantially reduce computational cost and enable fast prediction. However, most approaches still rely on high-fidelity FEM/FVM either to supply training data or to enforce physical constraints, and many are built on simplified heat-conduction or thermomechanical formulations, which bounds predictive capability by the physical completeness of the underlying models. Much of the literature focuses on pure heat conduction or simplified thermomechanical coupling, while melt-pool behavior is often approximated with equivalent heat sources, empirical correlations, or indirect treatments. Consequently, key mechanisms such as temperature-dependent surface tension, Marangoni effects, vapor recoil, and keyhole dynamics are seldom coupled explicitly at the surrogate level, limiting accuracy and generalization. Notably, purely data driven methods show strong potential in efficiency, modeling simplicity, and accuracy. With better physical priors and multiphysics consistency, such models could become more effective for complex melt-pool problems.

To address the trade-off among multiphysics consistency, cross-condition generalization, computational efficiency, and modeling complexity in existing thermo-fluid surrogates for AM and welding, inspired by prior work [42], [43], [44], [45], this study proposes a Physics-aware Autoregressive Encoder–Decoder Network (PAED-Net) that tightly integrates physical priors with temporal modeling. Concretely, PAED-Net employs a multi-scale convolutional encoder to extract spatial features from multi-physics fields (and velocity), and a skip-connected decoder to reconstruct full-field outputs, capturing both global and local features. A Physics-aware Subnet encodes

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material thermophysical properties, process inputs (e.g., laser power, scan speed, line energy density), and key dimensionless numbers as learnable priors injected into the main feature space, improving robustness across operating conditions. In addition, an adaptive, multi-physics loss with self-adjusted weights is used together with Efficient Channel Attention (ECA) to adaptively balance contributions across physical fields. In time, an autoregressive training strategy feeds the model's current prediction into the next step to realize long-horizon rollout, improving temporal stability with high efficiency.

The remainder of the paper is organized as follows. Section 2 presents the overall architecture of PAED-Net and the method in detail. Section 3 constructs a benchmark dataset for a canonical Marangoni case, evaluates accuracy, efficiency, and key ablations. Section 4 further validates the approach in laser welding and additive manufacturing by comparison with a high-fidelity simulation baseline, examining applicability and multiphysics consistency. Section 5 concludes the paper and discusses limitations and future work.

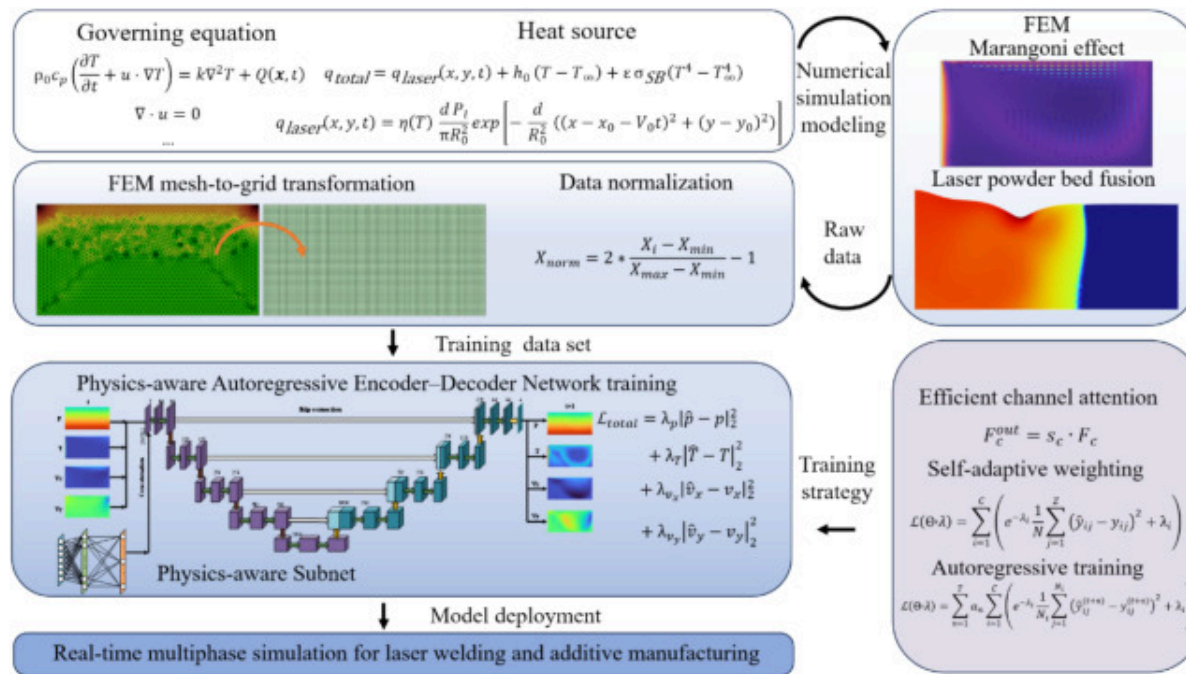
2. Methodology

2.1. Overall workflow

Fig. 1 illustrates the overall workflow of the proposed PAED-Net framework for real-time multiphysics simulation. First, the governing equations, heat source model, and boundary conditions for thermo-fluid coupling in the manufacturing process are established. High-fidelity FEM simulations are then performed to obtain time-resolved pressure, temperature, and velocity fields. The FEM data are mapped onto a regular grid and normalized to construct the training dataset. The PAED-Net model is trained using an autoregressive strategy with ECA and self-adaptive loss weighting to learn the spatio-temporal evolution of the coupled fields. A Physics-aware Subnet is introduced to al properties and process parameters as physical priors. Finally, the

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trained model is deployed for fast temporal rollout prediction of the multiphysics fields. The detailed architecture and training strategy are described in Section 2.3.



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Fig. 1. Overall framework of the PAED-Net for real-time multiphysics simulation.

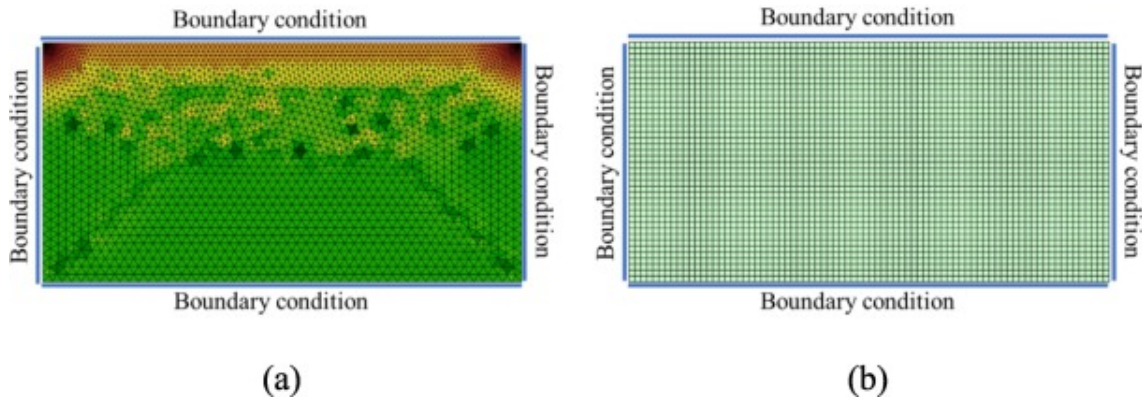
2.2. Data preprocessing

The FEM-computed multiphysics fields are defined on an unstructured mesh and cannot be directly used as network inputs. To address this, we first extract the spatial coordinates and the associated physical quantities from the numerical simulations, forming a set $\{(\mathbf{x}_k, \mathbf{y}_k, \mathbf{v}_k)\}_{k=1}^N$. The raw data are then filtered according to physical constraints to ensure physical consistency and interpretability. The fields are therefore remapped onto a regular two-dimensional grid, as shown in Fig. 2. Piecewise linear interpolation is used for grid mapping. Specifically, for any regular grid point $(\mathbf{x}_i, \mathbf{y}_j)$, located

inside a triangular $\triangle P_1 P_2 P_3$, its interpolated value is computed using barycentric coordinates as:

$$U^{(c)}(h, w) = \lambda_1 V^{(c)}(P_1) + \lambda_2 V^{(c)}(P_2) + \lambda_3 V^{(c)}(P_3) \quad (1)$$

where P_1, P_2, P_3 are the three vertices of the triangle, $V^{(c)}(P_m)$ denotes the value at vertex P_m , and $\lambda_1 + \lambda_2 + \lambda_3 = 1$. All physical fields are processed in this way and stacked along the channel dimension to form a regular-grid tensor $U \in \mathbb{R}^{H \times W \times C}$, where C is the number of physical fields and H and W denote the grid resolution. In this study, three grid resolutions are considered for the mesh-to-grid mapping, namely 64×128 , 128×256 , and 256×512 . These resolutions are further analyzed in the subsequent sections to evaluate the model performance under different spatial discretizations.



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Fig. 2. Mesh discretization and grid generation: (a) FEM mesh discretization; (b) corresponding uniform grid for the deep learning model.

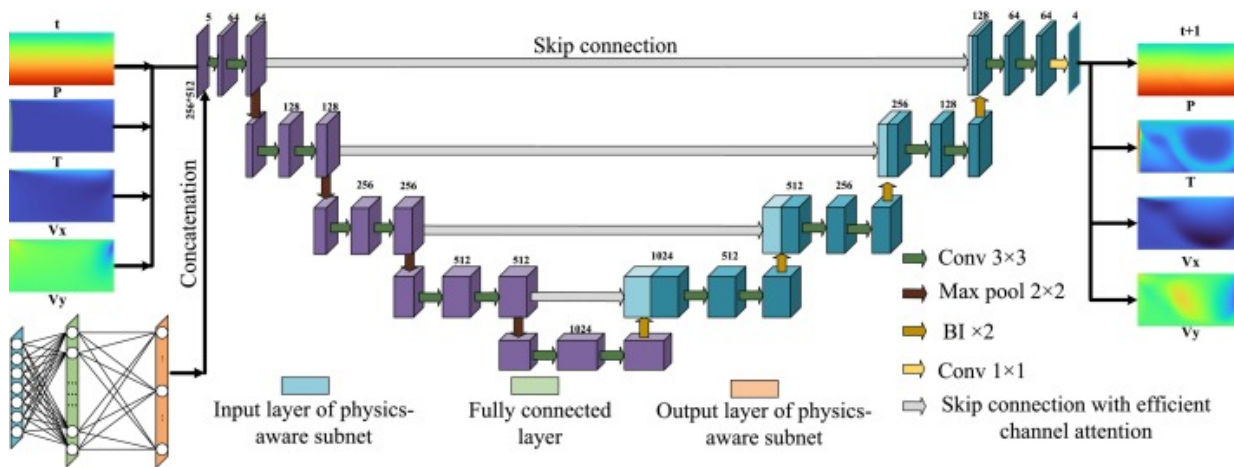
To reduce scale differences among physical fields, all inputs are linearly normalized to $[-1, 1]$. Specifically, for any physical quantity U_i , the value is first shifted by subtracting the minimum $U_{\min}$ and then scaled by the range $(U_{\max} - U_{\min})$ to obtain a quantity in $[0, 1]$. This intermediate value is further linearly mapped to

$[-1, 1]$, as given in Eq. (2):

$$U_{\text{norm}} = 2 * \frac{U_i - U_{\min}}{U_{\max} - U_{\min}} - 1 \quad (2)$$

2.3. PAED-Net

Fig. 3 shows the architecture of the proposed PAED-Net. The network consists of a physics-constrained autoregressive encoder–decoder backbone and a Physics-aware Subnet. The backbone adopts a multi-scale encoder–decoder with skip connections to process the stacked pressure, temperature and velocity fields, while an ECA module adaptively reweights different physical fields. The Physics-aware Subnet embeds material properties, boundary conditions and process parameters into compact learnable priors and fuses them with the backbone features, enabling controllable multiphysics prediction under varying operating conditions. An autoregressive training scheme is used to feed previous predictions back into the network for stable long-horizon forecasting. The detailed network configuration and training hyperparameters are summarized in Table 1. The backbone architecture adopts a U-Net–like encoder–decoder for spatial feature extraction, while the model is trained autoregressively to predict the physical state at the next time step from the current state.



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Fig. 3. PAED-Net architecture.

Table 1. Main parameters of PAED-Net.

Module	Layer / Operation	Configuration
Main structure	Downsampling	2D convolution, kernel size 3×3 , padding=1, stride=1; Max pooling with kernel 2×2 ; ReLU
	Upsampling	Bilinear interpolation with upsampling factor 2
	Output layer	2D convolution with kernel size 1×1
	ECA module	Global average pooling; 1D convolution with kernel size k ; Sigmoid activation
Physics-aware Subnet	Input layer	1×6 vector of material properties
	Fully connected	mapping $6 \rightarrow 256 \rightarrow 512$
	Output	Output feature tensor $U \in \mathbb{R}^{256 \times 512 \times 1}$

The network outputs physical fields with the same spatial resolution as the inputs, representing the state at time $T + 1$. The encoder consists of four downsampling blocks, each containing two DoubleConv units followed by Batch Normalization and a ReLU activation:

$$\mathbf{F}_i = \sigma(\text{BN}(\mathbf{W}_i * \mathbf{F}_{i-1} + \mathbf{b}_i))$$

(3)

where $\mathbf{F}_{i-1}$ denotes the feature map from the previous layer, $*$ is the convolution operation, $\mathbf{W}_i$ and $\mathbf{b}_i$ are learnable parameters, $\text{BN}(\cdot)$ denotes Batch Normalization, and σ is the sigmoid activation function. After each block, max pooling is applied to reduce the spatial

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resolution:

$$\mathbf{E}_i = \text{MaxPool}(\mathbf{F}_i) \quad (4)$$

where $\mathbf{E}_i$ is the pooled feature map.

The decoder progressively reconstructs the compressed global physical patterns obtained in the encoder and recovers them to the target-resolution multiphysics fields. Common upsampling strategies include bilinear interpolation, deconvolution, up-pooling, and patch expansion. Considering that the multiphysics fields in this work exhibit continuous and smooth spatial evolution, bilinear interpolation is adopted as the upsampling operator. In each upsampling block, the low-resolution feature maps are first upsampled using bilinear interpolation. The upsampled features are then fused with the corresponding high-resolution features from the encoder through skip connections. This design preserves both the global evolution patterns and the local boundary details of the multiphysics fields. The fused features are subsequently processed by a DoubleConv module followed by normalization and activation, yielding higher-resolution representations of the physical fields. The bilinear interpolation can be written as:

$$U_i = \sum_{m=0}^1 \sum_{n=0}^1 w_{mn} U_{i+1} \quad (5)$$

where U_{i+1} denotes the low-resolution input feature map, U_i is the unsampled output feature map, and w_{mn} are the bilinear interpolation weights. The upsampled feature map U_i is then concatenated along the physical field dimension with the corresponding encoder feature map U_i^{encoder} . This process can be expressed as:

$$\mathbf{L}_i^{\text{contact}} = \text{DoubleConv}(\text{Concat}(U_i, U_i^{\text{encoder}})) \quad (6)$$

where $\mathbf{L}_i^{\text{concat}}$ denotes the feature map after skip connection and convolutional fusion. This strategy helps the decoder recover fine-scale structures in the physical fields. At the same time, it maintains feature completeness and reduces the information loss

nsampling. The last feature map of the decoder is mapped to the target

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physical fields through a 1×1 convolution to obtain the final prediction:

$$\mathbf{U}_{\text{out}} = \mathbf{W}_{\text{out}} * \mathbf{L}_{\text{final}} + \mathbf{b}_{\text{out}} \quad (7)$$

where $\mathbf{L}_{\text{final}}$ is the final decoder feature map, $\mathbf{W}_{\text{out}}$ and $\mathbf{b}_{\text{out}}$ are the convolution kernel and bias of the output layer, and $\mathbf{U}_{\text{out}}$ denotes the predicted physical fields.

2.3.1. Efficient channel attention

In coupled multiphysics problems, different physical quantities often evolve on different spatial and temporal scales, especially near interfaces and source regions. In conventional simulations, resolving such fine-scale features usually requires local mesh refinement, which increases computational cost. After mapping the unstructured FEM mesh onto a uniform grid, all spatial locations are treated with similar importance, which may reduce the network's sensitivity to critical local regions. To address this, an ECA module is incorporated into the decoder, placed between the bilinear upsampling operation and the skip-connection concatenation with the corresponding encoder features [46]. ECA performs lightweight local cross-channel interactions to adaptively learn the importance of different physical fields. It suppresses redundant information while enhancing the response to salient physical variation patterns. Let the input feature map be $\mathbf{F} \in \mathbf{R}^{C \times H \times W}$. Global average pooling is first applied to obtain a compact channel-wise descriptor:

$$\mathbf{z}_c = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W \mathbf{F}_c(i, j) \quad (8)$$

where $\mathbf{z}_c$ denotes the global descriptor of the c -th physical field. Then, a one-dimensional convolution with kernel size $\mathbf{K}$ is applied along the physical field dimension, followed by a Sigmoid activation to generate the physical field attention weights:

$$\mathbf{e} = \sigma(\text{Conv1D}(\mathbf{z})) \quad (9)$$

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where $\text{Conv1D}(\cdot)$ denotes the 1D convolution and $\sigma(\cdot)$ is the Sigmoid function. The kernel

size K is adaptively determined as a function of the physical field number C to control the range of local cross-channel interactions.

$$K = \frac{\log_2 D + 1}{2} \quad (10)$$

Finally, the obtained weights are used to rescale each channel of the original feature map:

$$\mathbf{F}_c^{\text{out}} = \mathbf{e}_c \bullet \mathbf{F}_c \quad (11)$$

where $\mathbf{F}_c^{\text{out}}$ is the reweighted feature of channel c . This channel-wise recalibration strengthens informative physical patterns while mitigating the loss of critical information caused by uniform spatial weighting.

2.3.2. Self-adaptive weighting loss function

For the four primary physical fields considered in this work—pressure p , temperature T , horizontal velocity v_x , and vertical velocity v_y , the total loss is defined as:

$$\mathcal{L}_{\text{total}} = \lambda_p \|\hat{p} - p\|_2^2 + \lambda_T \|\hat{T} - T\|_2^2 + \lambda_{v_x} \|\hat{v}_x - v_x\|_2^2 + \lambda_{v_y} \|\hat{v}_y - v_y\|_2^2 \quad (12)$$

where $\hat{p}, \hat{T}, \hat{v}_x, \hat{v}_y$ denote the predictions, p, T, v_x, v_y are the corresponding ground truth fields, $\|\cdot\|_2^2$ denotes the mean squared error (MSE), and $\lambda_p, \lambda_T, \lambda_{v_x}, \lambda_{v_y}$ are the loss weights for the four channels. In the laser welding and additive manufacturing case, the same loss formulation is extended to five channels by additionally including the volume-fraction field V_f . Although these physical fields are strongly coupled, their units and numerical scales differ significantly. Using a shared weight may cause large-scale channels to dominate the optimization and degrade overall accuracy and stability. To ensure accurate prediction across all fields, we adopt an uncertainty-based multi-task loss design and introduce a self-adaptive channel weighting scheme [47]. Specifically, a learnable log-variance parameter s_c is assigned to each field, and e^{-s_c} is used as its adaptive weight to mitigate scale imbalance. The weighted sum of per-field MSEs is used

as the optimization objective:

$$\mathcal{L}(\Theta, \mathbf{s}) = \sum_{c=1}^C \left(e^{-s_c} \frac{1}{N_c} \sum_{j=1}^{N_c} (\hat{y}_{cj} - y_{cj})^2 + s_c \right) \quad (13)$$

where Θ denotes the model parameters, s_c is the log-variance parameter for the c -th field, N_c is the number of samples for that field, and $\hat{y}_{cj}$ and y_{cj} are the predicted and true values of channel c at sample j , respectively. Accordingly, the weights in Eq. (12) can be written as: $\lambda_p = e^{-s_p}$, $\lambda_T = e^{-s_T}$, $\lambda_{v_x} = e^{-s_{v_x}}$, $\lambda_{v_y} = e^{-s_{v_y}}$.

2.3.3. Autoregression training

To improve temporal consistency and cross-step generalization, we employ a fully autoregressive multi-step rollout strategy during training [48]. Within a rolling window of length T , the prediction at each step is fed back as the input for the next step, forming an end-to-end temporal chain that aligns the input distributions of training and inference (see Fig. 4). The autoregressive window length determines the number of consecutive time steps involved in the multi-step rollout during training. Its influence on model performance is further analyzed in Section 3.4. Let U_0 denote the initial input field, $\mathbf{q}$ the vector of controllable attributes (e.g., material and process parameters), and $f_{\Theta}(\cdot)$ the PAED-Net model. The prediction at time step t is given by:

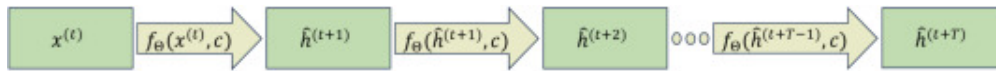
$$\hat{U}_{t+1} = f_{\Theta}(\hat{U}_t, \mathbf{q}) \quad (14)$$

where $t = 1, 2, \dots, T$, with $\hat{U}_1 := U_1$. Denote $\hat{y}_{cj} = \hat{U}_{t+1}^{(c)}[j]$, and $y_{cj} = U_{t+1}^{(c)}[j]$, as the predicted and true values at time $t + 1$, channel c , and pixel/grid point j . At each time step and for each channel, the pixel-wise MSE is computed and weighted by e^{-s_c} .

Summing over all time steps yields the temporally extended autoregressive loss:

$$\mathcal{L}_{AR} = \sum_{t=1}^{T-1} \sum_{c=1}^C \left[e^{-s_c} \frac{1}{N_c} \sum_{j=1}^{N_c} (\hat{U}_{t+1}^{(c)}[j] - U_{t+1}^{(c)}[j])^2 \right] + \sum_{c=1}^C s_c. \quad (15)$$

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Fig. 4. Autoregressive Training Flowchart.

In this way, channels with larger residuals receive higher weights along the temporal dimension, while noisy channels are moderately down-weighted, leading to improved numerical stability and physical consistency across both time steps and multiple physical fields. In summary, Eq. (12) defines the basic multi-field MSE loss, Eq. (13) introduces uncertainty-based self-adaptive channel weighting, and Eq. (15) represents its autoregressive extension along the temporal dimension.

2.3.4. Physics-aware subnet

High-fidelity numerical simulations rely on accurate external conditions such as material properties and process parameters. To explicitly inject these physical priors into the network, we design a Physics-aware Subnet that embeds external variables (e.g., material and process parameters) into the feature space. It should be noted that the Physics-aware Subnet mainly provides global conditioning information rather than explicitly specifying the spatial position of the heat source. In the LPBF case, the location of the moving laser is implicitly encoded in the multiphysics fields at the current time step (e.g., temperature, velocity, pressure, and volume fraction), which already reflect the spatial response of the melt pool to the current laser position. The parameter vector is mapped into a learnable two-dimensional conditioning map and fused with these field inputs, allowing the network to combine global process conditions with the evolving spatial state when predicting the next time step. Let $\mathbf{m} = [m_1, m_2, \dots, m_d] \in \mathbb{R}^d$ denote the low-dimensional vector of physical attributes. A fully connected layer maps $\mathbf{m}$ to a 2D conditioning map $M \in \mathbb{R}^{H \times W}$ with the same spatial resolution as the backbone

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feature maps:

$$\text{vec}(\mathbf{m}) = \mathbf{W}\mathbf{m} + \mathbf{b} \quad (16)$$

where $\text{vec}(\cdot)$ denotes vectorization, $\mathbf{W} \in \mathbb{R}^{(HW) \times d}$ is the weight matrix, $\mathbf{b} \in \mathbb{R}^{HW}$ is the bias term, and H and W are the height and width of the spatial feature maps, respectively. The resulting 2D attribute map $\mathbf{M}$ is then treated as an additional channel and concatenated with the original spatial features along the channel dimension:

$$\mathbf{X}_{\text{input}} = \text{Concat}(\mathbf{X}, \mathbf{M}), \quad (17)$$

so that the encoder can jointly learn from spatial fields and operating conditions.

2.4. Evaluation metrics

To quantitatively evaluate the prediction performance of the model, four error metrics are employed: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Max Error, and Min Error. For the c -th physical field, they are defined as:

$$\text{RMSE}^{(c)} = \sqrt{\frac{1}{H * W} \sum_{w=1}^W \sum_{h=1}^H \left(\hat{Y}_{hw}^{(c)} - Y_{hw}^{(c)} \right)^2}, \quad (18)$$

$$\text{MAE}^{(c)} = \frac{1}{H * W} \sum_{w=1}^W \sum_{h=1}^H \left| \hat{Y}_{hw}^{(c)} - Y_{hw}^{(c)} \right|, \quad (19)$$

$$\text{MaxError}^{(c)} = \max_{hw} \left| \hat{Y}_{hw}^{(c)} - Y_{hw}^{(c)} \right|, \quad (20)$$

$$\text{MinError}^{(c)} = \min_{hw} \left| \hat{Y}_{hw}^{(c)} - Y_{hw}^{(c)} \right|. \quad (21)$$

Here, $\text{RMSE}^{(c)}$, $\text{MAE}^{(c)}$, $\text{MaxError}^{(c)}$, and $\text{MinError}^{(c)}$ denote the root mean squared error, mean absolute error, maximum error, and minimum error for the c -th physical

Typesetting math: 100% $\hat{Y}_{hw}^{(c)}$ and $Y_{hw}^{(c)}$ are the predicted and ground-truth values at spatial

location (h, w) , and $\max_{h,w}$ and $\min_{h,w}$ indicate the maximum and minimum errors over all spatial locations. Here, physical fields denote the underlying variables (P, T, u, v, V_f) . Channels denote the tensor dimension in the network input/output, with a one-to-one correspondence to the physical fields; hence C equals the number of fields (e.g., $C = 4$ for the Marangoni case and $C = 5$ for Laser welding and additive manufacturing).

3. Case study in Marangoni effect

The Marangoni effect is driven by temperature-induced surface-tension gradients along the free surface, which generate tangential shear stress and reshape the internal flow. It affects heat and mass transport, microstructure evolution, and solute distribution, and is therefore a key factor in additive manufacturing quality. Since this work combines high-fidelity simulations with deep learning, a large number of equations and symbols are involved. For readability, some notation is reused across sections, with its meaning clarified by context.

3.1. Numerical simulation of Marangoni effect

For readability, some notation is reused across sections, with its meaning clarified by context. To isolate the Marangoni-dominated mechanism, the following assumptions are adopted: (1) the liquid is incompressible and its volume does not change with pressure; (2) the fluid is Newtonian with a constant dynamic viscosity μ ; and (3) the Boussinesq approximation is applied, i.e., the density is treated as constant in the continuity equation and allowed to vary linearly with temperature only in the buoyancy term. The governing equations are:

Mass conservation:

$$\nabla \cdot \mathbf{u} = 0, \quad (22)$$

Momentum equation:

$$\rho_0 \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \mu \nabla^2 \mathbf{u} + \rho_0 \beta (T - T_0) \mathbf{g}, \quad (23)$$

Energy equation:

$$\rho_0 c_p \left(\frac{\partial T}{\partial t} + \mathbf{u} \cdot \nabla T \right) = k \nabla^2 T + Q(\mathbf{x}, t), \quad (24)$$

where $\mathbf{u}$ is the velocity field, ρ_0 is the reference density, $\partial \mathbf{u} / \partial t$ is the local unsteady acceleration, $\mathbf{u} \cdot \nabla \mathbf{u}$ represents the convective acceleration, $-\nabla p$ is the pressure gradient, $\rho_0 \beta (T - T_0) \mathbf{g}$ is the buoyancy force under the Boussinesq approximation, and $\mu \nabla^2 \mathbf{u}$ accounts for viscous diffusion. In the energy equation, $\rho_0 c_p$ is the volumetric heat capacity, $\partial T / \partial t$ is the transient term, $\mathbf{u} \cdot \nabla T$ corresponds to convective heat transfer, $k \nabla^2 T$ is the heat conduction term with thermal conductivity k , and $Q(\mathbf{x}, t)$ denotes the volumetric heat source. The Marangoni effect at the free surface is imposed through a tangential stress boundary condition rather than a body force. On the top boundary Γ_s , the tangential stress condition is prescribed as:

$$\mathbf{t} \cdot \left[\mu \left(\nabla \mathbf{u} + (\nabla \mathbf{u})^\top \right) \right] \cdot \mathbf{n} = \frac{\partial \sigma}{\partial T} \nabla_s T \cdot \mathbf{t} \quad (25)$$

where ∇_s is the surface gradient operator, and $\mathbf{n}$ and $\mathbf{t}$ are the unit normal and tangential vectors on the free surface, respectively. The geometric deformation of the free surface is not explicitly tracked; instead, a flat free surface with a no-penetration condition is assumed.

The computational domain is a two-dimensional rectangular region with a length of 10mm and a height of 5mm. This aspect ratio is sufficient to capture the main recirculation and shear layer. An unstructured triangular mesh is generated automatically, with a maximum element size of 0.2mm and a minimum element size of 0.00075mm. To balance accuracy and efficiency, the mesh is refined near the free surface and side walls,

sh is used in the interior. To evaluate the applicability and generalization capability of the model across different materials, three representative

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liquid systems are considered: silicone oil, water, and ethylene glycol. Because data-driven learning is sensitive to numerical scales, large differences in units and magnitudes may cause gradient imbalance and unstable convergence. For data preprocessing and neural-network training, the physical properties listed in Table 4 are linearly scaled to avoid large differences in numerical magnitude; the underlying CFD simulations are still performed with the original, unscaled properties. Table 2 summarizes the temperature coefficient of surface tension ($-\gamma$), fluid density (ρ), dynamic viscosity (μ), thermal conductivity (κ), specific heat capacity (c_p), and thermal expansion coefficient (β) for silicone oil, water, and ethylene glycol, together with their scaled values, to facilitate cross-material comparison.

Table 2. Physical properties of silicone oil, water, and ethylene glycol.

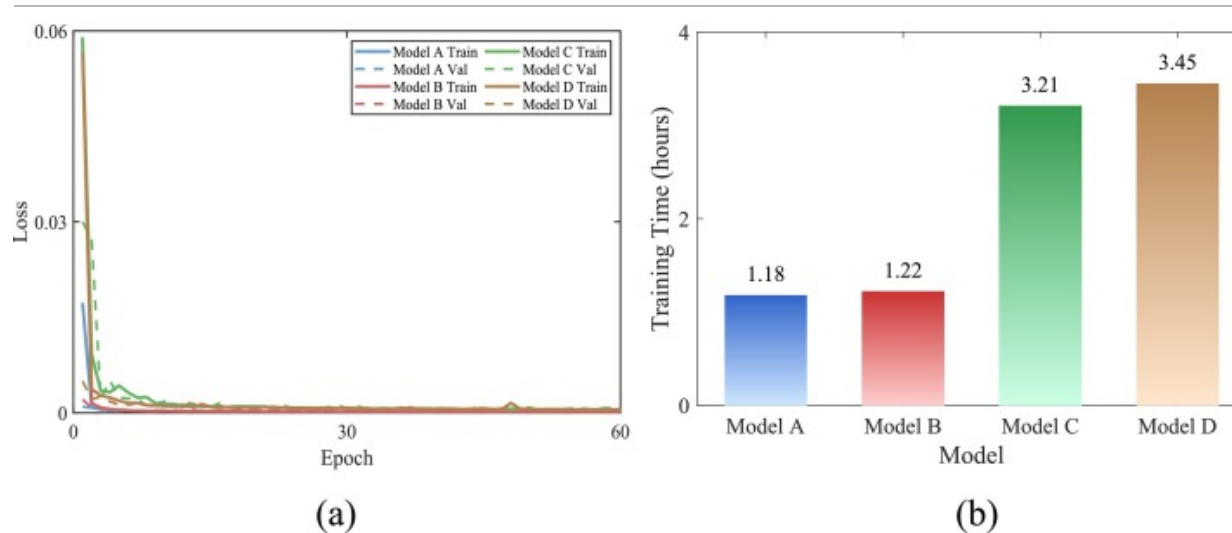
Material	$-\gamma$ [N/(m · K)]	ρ [kg /m ³]	μ [Pa · s]	κ [W/(m · K)]	c_p [J/(kg · K)]	β [1/K]
Silicone oil	$-8e^{-5}$	760	$4.94e^{-4}$	0.1	2090	0.0013
	−0.08	0.076	0.00494	0.1	0.209	0.0013
Water	$-1.5e^{-4}$	1000	0.001	0.6	4200	$2e^{-4}$
	−0.15	0.100	0.01	0.6	0.420	0.0002
Ethylene glycol	$-1.7e^{-4}$	1113.2	0.0161	0.258	2420	$5.7e^{-4}$
	−0.17	0.11132	0.161	0.258	0.242	0.00057

3.2. Model training

To evaluate the impact of different architectural components on multiphysics prediction, four comparative variants are constructed on top of a unified encoder–decoder backbone: Model A (baseline), Model B (+ECA), Model C (+autoregressive training), and Model D (+ECA +autoregressive training). In total, 2760 pairs of samples are generated for silicone oil, water, and ethylene glycol (920 pairs for each material). Unless otherwise stated, 80%

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of the samples are used for training and the remaining 20% for validation. All models are trained on a workstation equipped with an NVIDIA GeForce RTX 3060 GPU (12GB), an Intel Core i5-14600KF CPU, 32GB of RAM, CUDA 12.2, and PyTorch 2.3. As shown in Fig. 5(a), the training loss of all four models decreases rapidly within the first 10 epochs and becomes nearly stable after about 20 epochs. Due to the backpropagation of temporal rollout errors introduced by autoregressive training, Model C exhibits a higher overall loss level and slower convergence. After incorporating ECA, the early-stage convergence lag of Model D is partially alleviated, but its convergence remains slower than that of Model A and Model B, mainly because multi-step prediction errors are accumulated and backpropagated along the temporal dimension during autoregressive training. Fig. 5(b) compares the training time of the four models: the training times of Model A and Model B are approximately 1.18h and 1.22h, respectively, whereas Model C and Model D, which adopt the autoregressive training strategy, require 3.21h and 3.46h. As indicated by Eq. (15), autoregressive training entails evaluating the loss along the temporal dimension and backpropagating through multiple time steps, thereby incurring a substantially higher computational cost, while the additional overhead introduced by the ECA module is relatively minor.

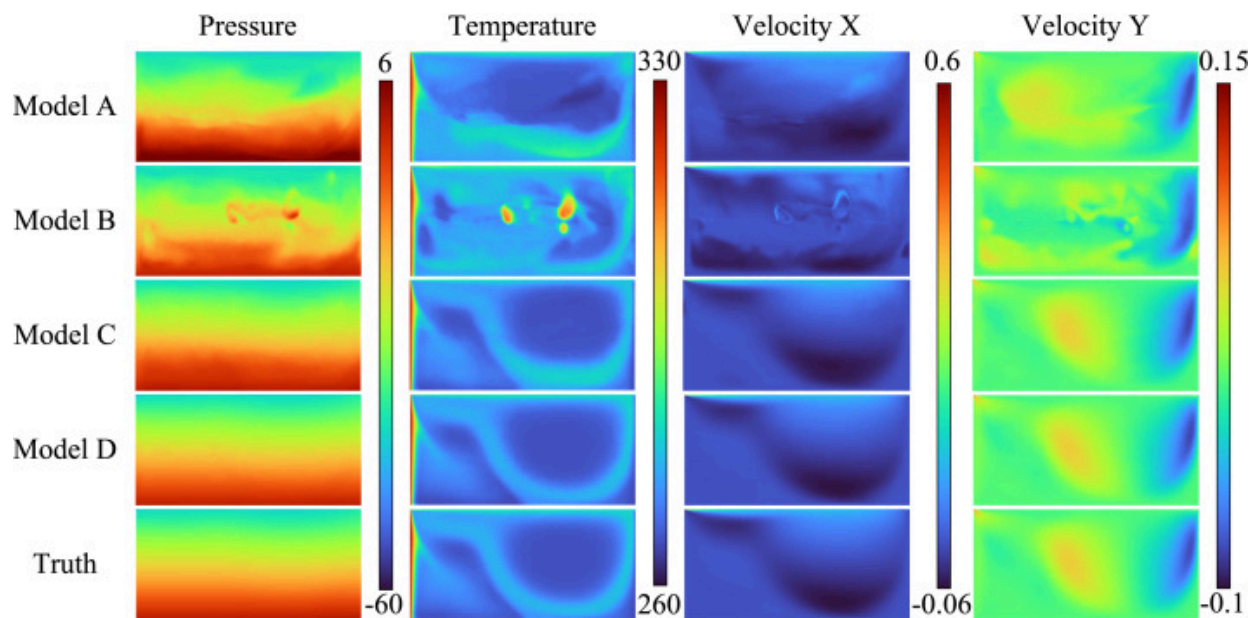


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Fig. 5. Comparison of the training performance of the four models: (a) training loss curves; (b) training time.

3.3. Comparison of models

To objectively evaluate the performance of different architectures in temporal rollout prediction, the time domain is discretized with a fixed time step as $t = 1, \dots, T - 1$, with $T = 40$ chosen as the overall prediction horizon. At each time step, the model output from the previous step is fed back as the input for the next step (consistent with the autoregressive setting in Section 3.2). Qualitative comparisons are conducted at the 40th time step, while quantitative metrics are evaluated over all 40 time steps. As shown in Fig. 6, Model A reproduces the overall distribution patterns of each physical field, but its response to high-gradient regions and fine-scale structures is insufficient. Model B, which augments Model A with ECA, exhibits improved sensitivity to local features; however, spurious “hot spots” appear in the temperature field, and non-physical perturbations and oscillations are observed in the velocity field, indicating that under single-step training, the attention mechanism alone is not sufficient to suppress the accumulation of multi-step rollout errors. By introducing autoregressive training, Model C explicitly models temporal dependence, leading to a marked improvement in cross-step consistency: it preserves the global evolution trends of pressure and velocity while more accurately capturing high-gradient bands. Model D, which combines autoregressive training with ECA, further enhances local detail reconstruction and, compared with Model B (attention only), achieves more stable spatial field quality and better temporal consistency.

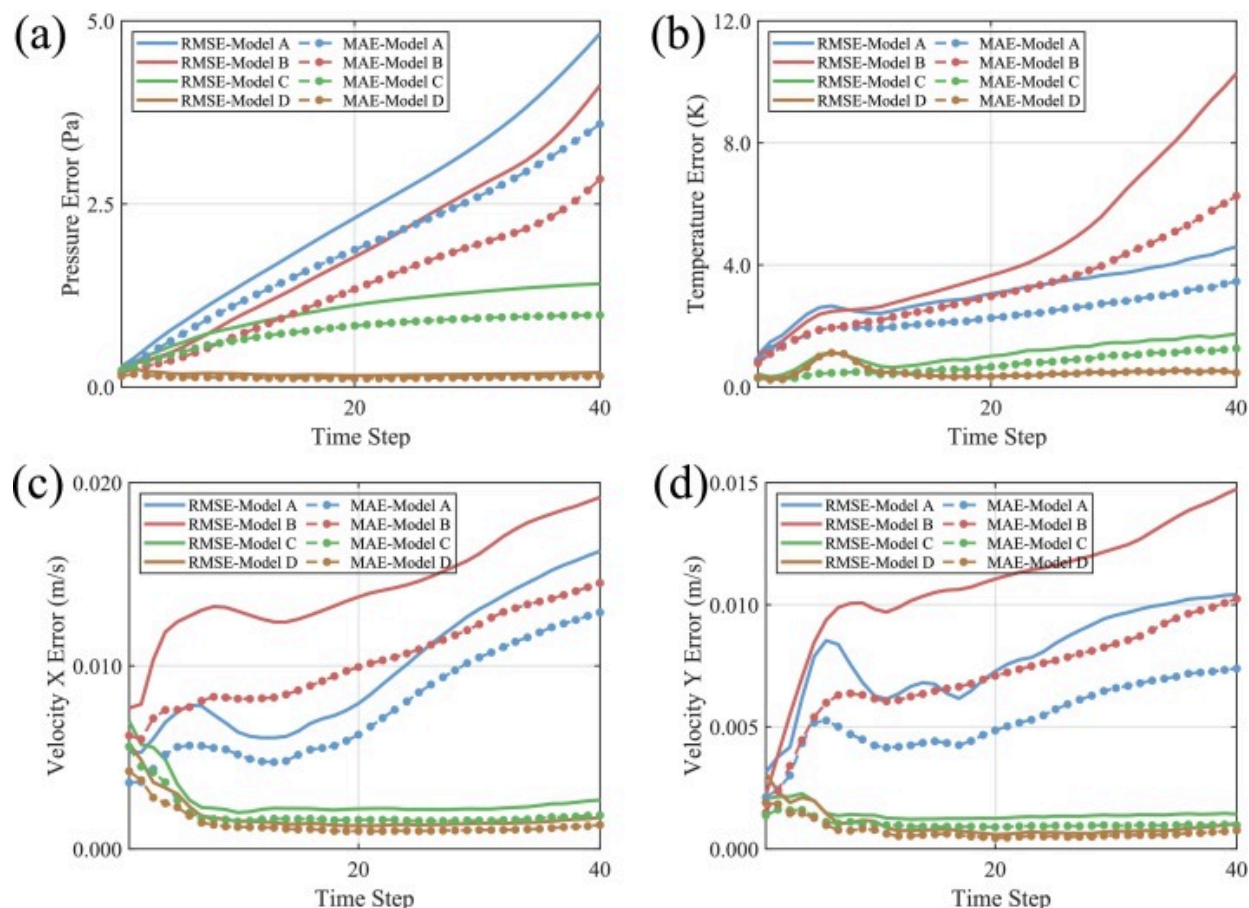


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Fig. 6. Comparison between the predictions of different models and the FEM reference solutions.

Fig. 7 shows the evolution of RMSE and MAE over 40 rollout steps for the four models. For Models A and B, the errors increase almost monotonically with the time step, with a pronounced error diffusion in the later stages. Model B performs slightly better than Model A for the pressure field. However, the differences in RMSE and MAE among different physical fields become larger. Model C exhibits lower and more stable error curves, indicating improved temporal consistency. Building on this, Model D further reduces the errors, and its RMSE and MAE curves almost overlap, reflecting higher stability and accuracy. This indicates that autoregressive training improves rollout stability. It is also observed that the benefit of ECA becomes more evident when it is combined with the autoregressive strategy. In contrast, using ECA alone does not the accumulation of multi-step prediction errors.



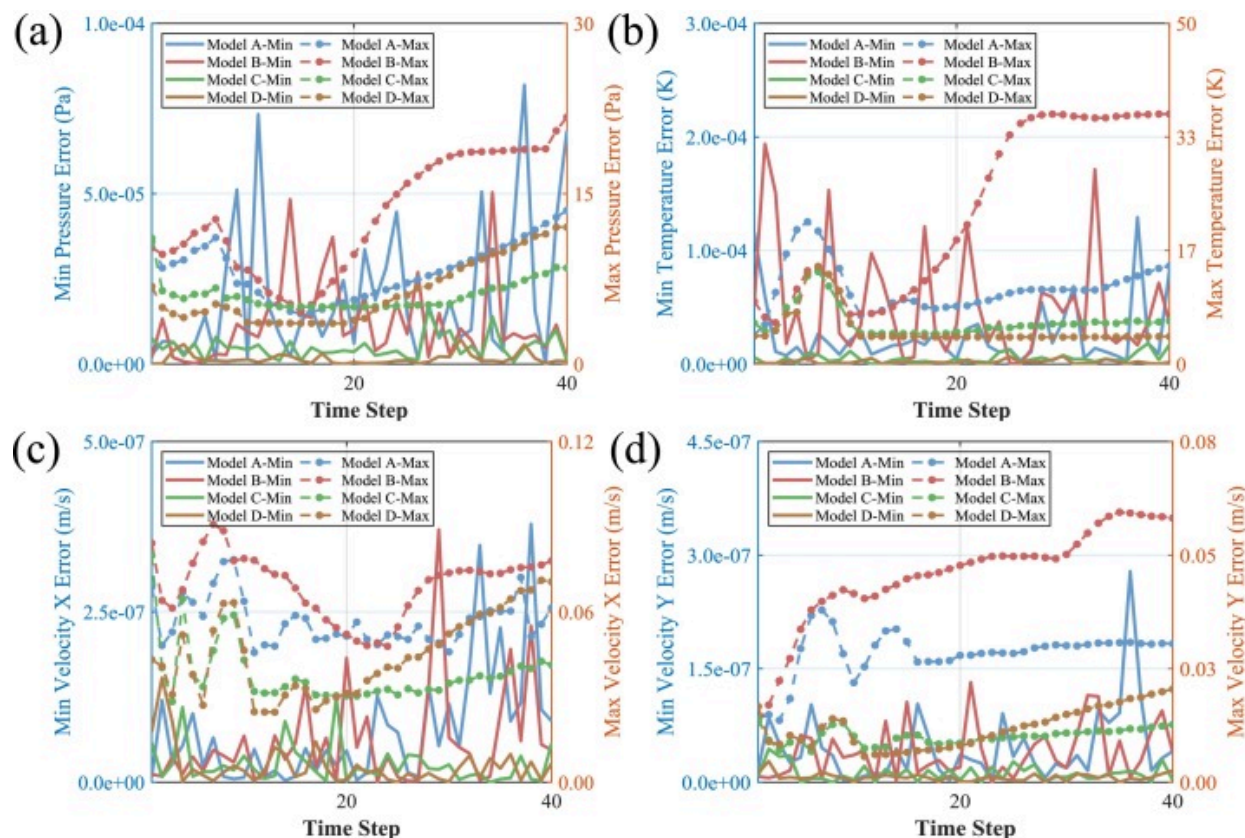
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Fig. 7. Evolution of RMSE and MAE over 40 rollout steps for the four models: (a) Pressure error; (b) Temperature error; (c) Velocity X error; (d) Velocity Y error.

Fig. 8 shows the evolution of the extreme errors over 40 time steps for the four models. For the pressure field, the minimum error remains close to zero across the entire time domain, with only a few spikes, whereas the maximum error increases with time; Model B exhibits a faster rise and larger peaks after about the 14th step, while Models C and D show a slower growth and markedly reduced maxima. In the temperature field, the

minimum error stays low and its sporadic spikes are consistent with the spurious hot spots observed in Fig. 6. The maximum error of Models A and B increases in a stepwise manner from roughly the 15th step and remains high, whereas Models C and D maintain a low and smooth profile, indicating a more robust treatment of the high-gradient region near the free surface. For the velocity components v_x and v_y , the minimum error is low and stable for all four models, but the maximum error in Models A and B grows significantly over time with multiple pronounced spikes. In contrast, Models C and D keep the maximum error at low amplitudes with small temporal fluctuations. Together with the isotherm and streamline comparisons in Fig. 6, these results indicate that the autoregressively trained Models C and D effectively mitigate multi-step error accumulation, with Model D achieving the best overall temporal stability and physical consistency.



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Fig. 8. Evolution of the maximum and minimum prediction errors over 40 time steps for the four models: (a) Pressure error; (b) Temperature error; (c) Velocity X error; (d) Velocity Y error.

For a more intuitive comparison, the RMSE of each physical quantity is first averaged over all 40 time steps and then normalized by the dynamic range of that quantity on the evaluation set, yielding a dimensionless relative error in percentage. As shown in Table 3, Model D achieves the smallest relative error for all four physical quantities, followed by Model B, which only adds ECA, does not show a clear advantage. The

rankings given by the relative errors are consistent with the RMSE, MAE, and extreme-error analyses. Overall, autoregressive training is the main factor improving multi-step stability, while ECA provides additional gains when combined with temporal rollout training.

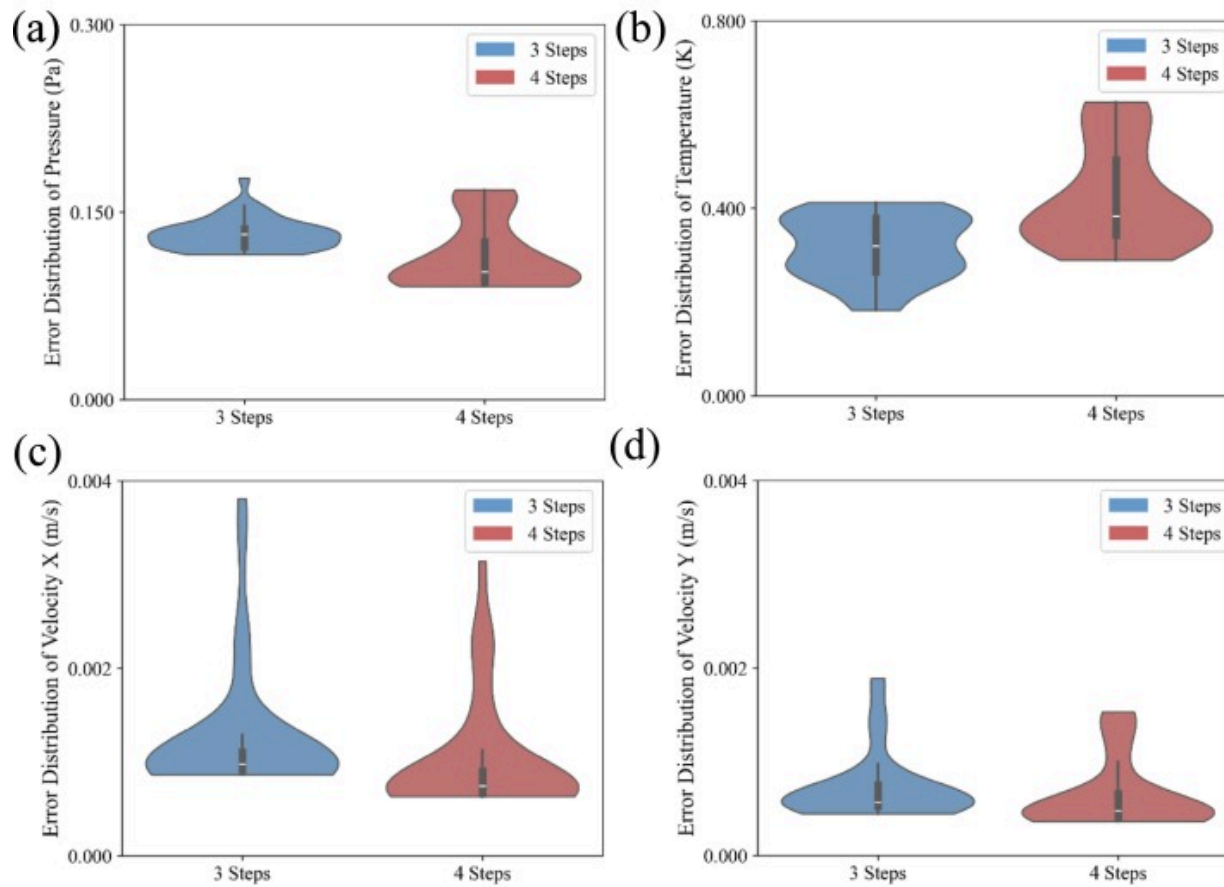
Table 3. Relative errors (%) of the four models for different physical quantities.

Physical fields	Model A	Model B	Model C	Model D
P	4.48%	3.54%	1.93%	0.33%
T	5.80%	8.47%	2.06%	0.92%
V_x	2.62%	3.83%	0.70%	0.47%
V_y	5.79%	7.98%	1.06%	0.73%

3.4. Analysis of autoregressive steps

To assess the effect of autoregressive window length on multi-step rollout prediction, Model D is evaluated using 3-step and 4-step configurations with the same architecture and hyperparameters. The error distributions of the four physical fields are shown as violin plots in Fig. 9. For the pressure and velocity fields, the 4-step configuration yields lower median errors and more compact distributions, indicating that a longer temporal window better constrains the Marangoni-shear-driven surface jet and recirculation, and reduces local deviations near the shear layer and recirculation region. In contrast, for the temperature field, the 3-step configuration shows a lower median error and lighter tails, whereas the 4-step configuration exhibits a broader spread and heavier tails, suggesting that surface temperature is more prone to error accumulation and drift over longer rollouts. Thus, extending the window does not uniformly benefit all physical fields. The training times for 3-step and 4-step autoregression are approximately 3.45h and 6.20h, the increased memory cost makes longer rollouts difficult on the current hardware.

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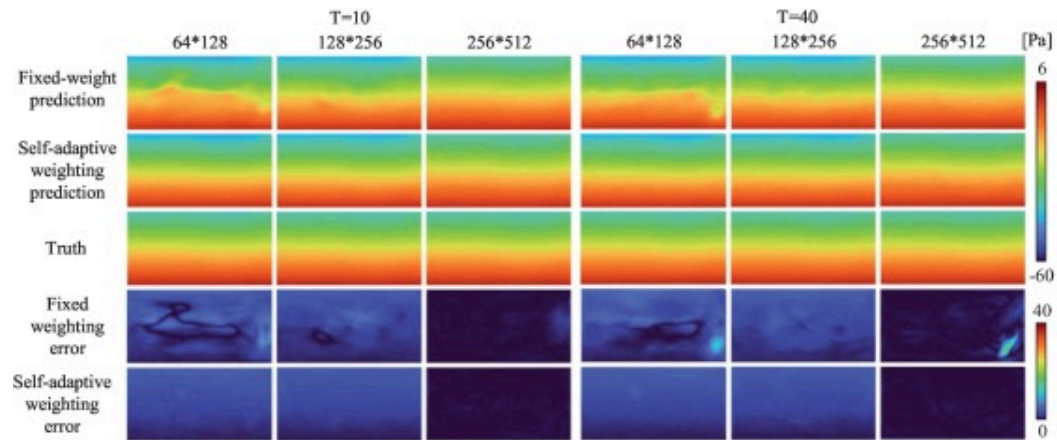
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Fig. 9. Violin plots of error distributions for different autoregressive step numbers: (a) Error distribution of pressure; (b) Error distribution of temperature; (c) Error distribution of velocity X; (d) Error distribution of velocity Y.

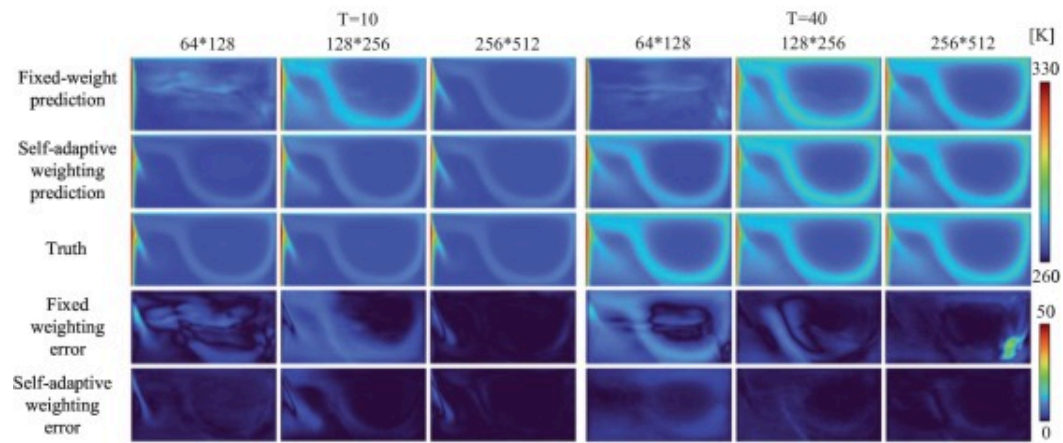
3.5. Analysis of self-adaptive weighting loss function

To assess robustness across spatial discretizations, predictions are performed at three resolutions: 64×64 , 128×128 , and 256×256 . When the resolution is reduced to 64×128 , the prediction accuracy deteriorates significantly compared with the higher-resolution

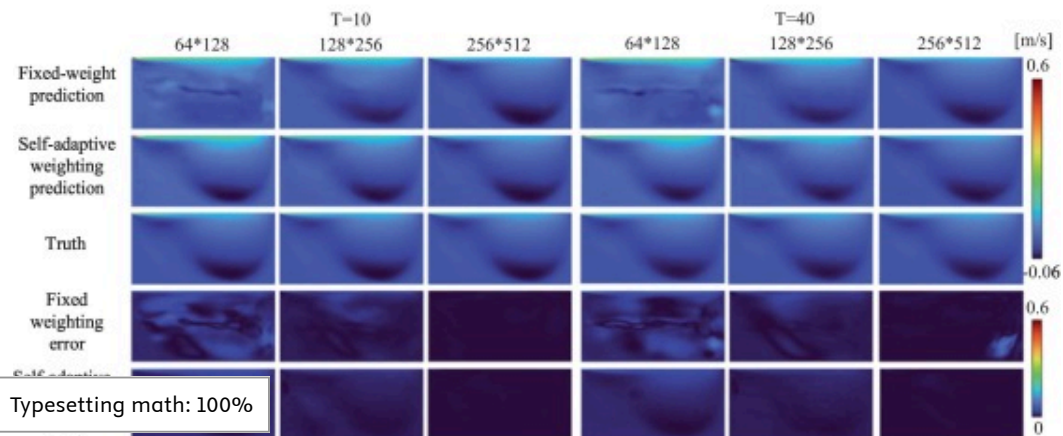
cases. This is mainly because low-resolution inputs contain less physical information and alter the spatial patterns of multiphysics fields, which disrupts the balance among channels in the loss function and reduces the effectiveness of fixed loss weights. To address this issue, the self-adaptive weighting strategy introduced in Section 2.3.2 is employed, allowing the loss weights of different physical fields to be dynamically adjusted with changing data scale and resolution. Fig. 10 compares the fixed and self-adaptive loss-weight configurations at the 10th and 40th time steps for the three spatial resolutions 64×128 , 128×256 , and 256×512 . In each column, the rows from top to bottom correspond to the fixed-weight prediction, the self-adaptive prediction, the FEM ground truth, and the error map. In Fig. 10(a), the pressure field obtained with fixed loss weights exhibits pronounced strip-like fluctuations, which become more evident at low resolution and at the 40th time step. With self-adaptive weighting, an almost linear hydrostatic pressure profile and a smoother transition near the free surface are recovered. In Fig. 10(b), the temperature field under fixed loss weights develops isolated hot spots at low resolution and long prediction horizons, whereas self-adaptive weighting better preserves the thermal boundary layer and isotherm curvature. Fig. 10(c) shows that, for v_x , fixed loss weights lead to stripe-like noise and shear-layer misalignment, whereas self-adaptive weights allow a continuous surface jet and downstream recirculation wake to be reconstructed even at 64×128 . Finally, Fig. 10(d) indicates that, for v_y , fixed loss weights at low resolution cause the vortex core of the main recirculation cell to deviate from the reference position, while self-adaptive weighting maintains a more stable vortex position and strength.



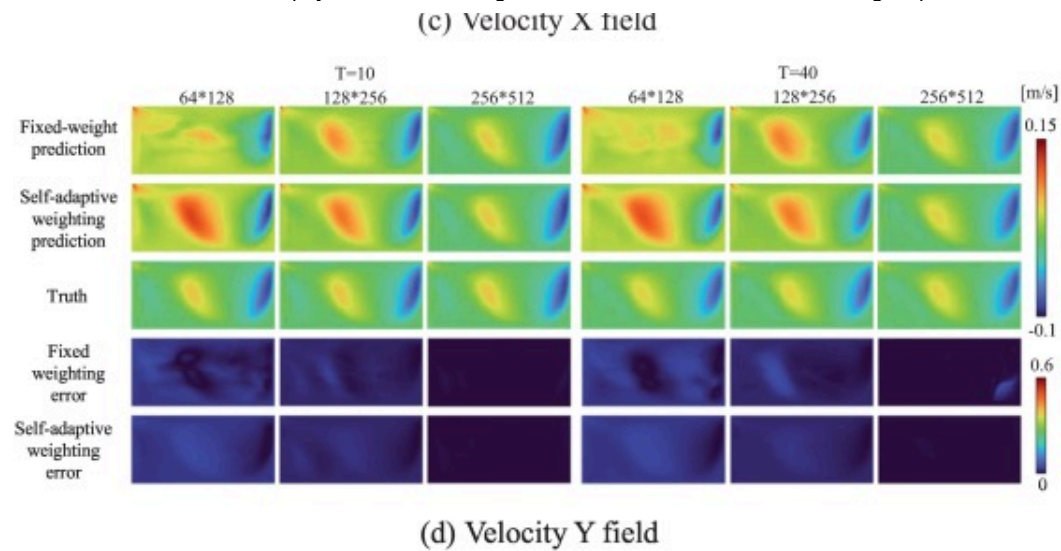
(a) Pressure field



(b) Temperature field



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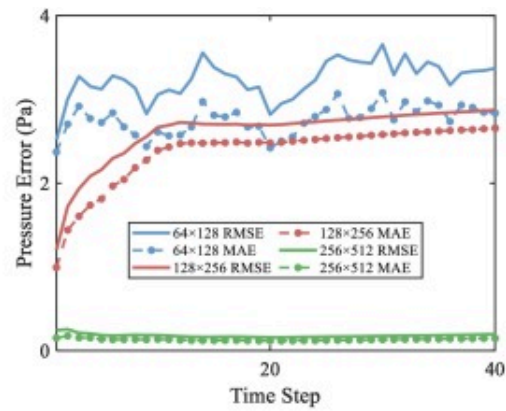
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Fig. 10. Predictions with self-adaptive vs. fixed weights at varied resolutions.

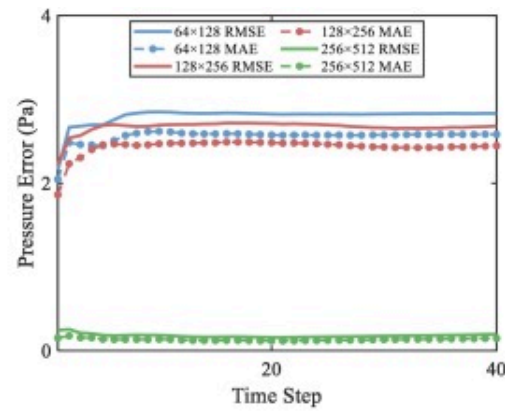
Fig. 11 compares the evolution of RMSE and MAE for the configurations with fixed and self-adaptive loss weights. For the pressure field, RMSE and MAE under fixed loss weights increase with time and exhibit pronounced spikes, whereas self-adaptive loss weights reduce the error level and lead to earlier convergence. For the temperature field, fixed loss weights combined with low spatial resolution cause continuous error growth, while self-adaptive loss weights substantially alleviate this effect and rapidly reduce the error to a low and stable level at higher resolutions, indicating a more robust representation of the high-gradient region near the free surface. For the velocity fields, fixed loss weights yield persistently higher average errors and a more noticeable gap between RMSE and MAE, suggesting a heavier-tailed error distribution. In contrast, self-adaptive loss weights lower the error curves and smooth their fluctuations, although a slight increase is still observed near the end of the rollout at the highest resolution, likely due to phase drift of small-
the shear layer and recirculation region. Overall, increasing spatial resolution reduces RMSE and MAE and suppresses temporal oscillations, while self-

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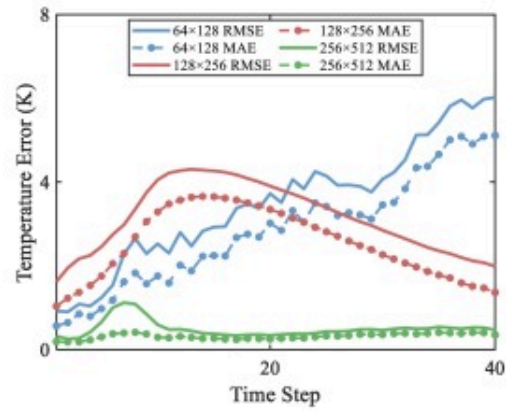
adaptive loss weights further decrease the average errors and mitigate error growth around time steps 20–40.



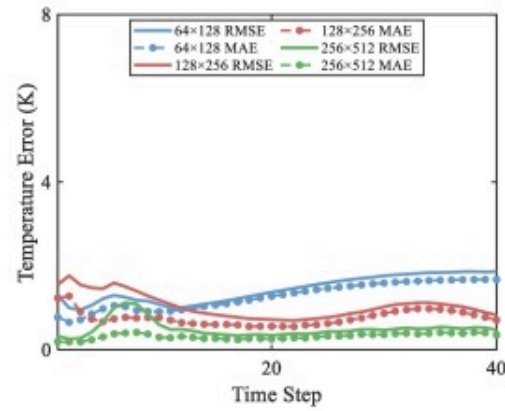
(a)



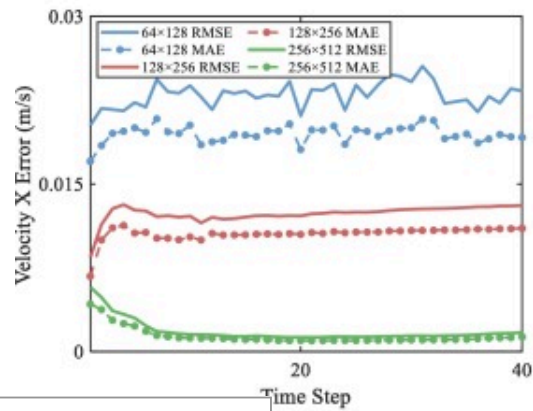
(b)



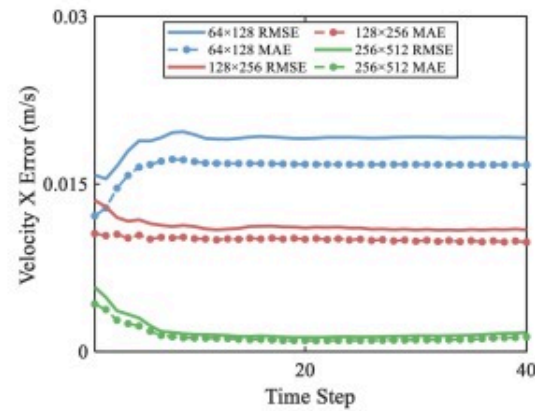
(c)



(d)

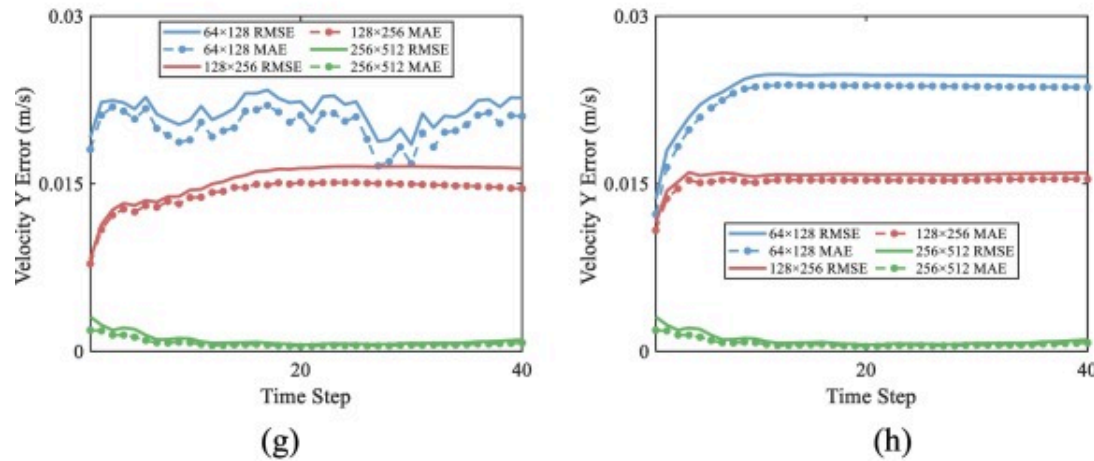


(e)



(f)

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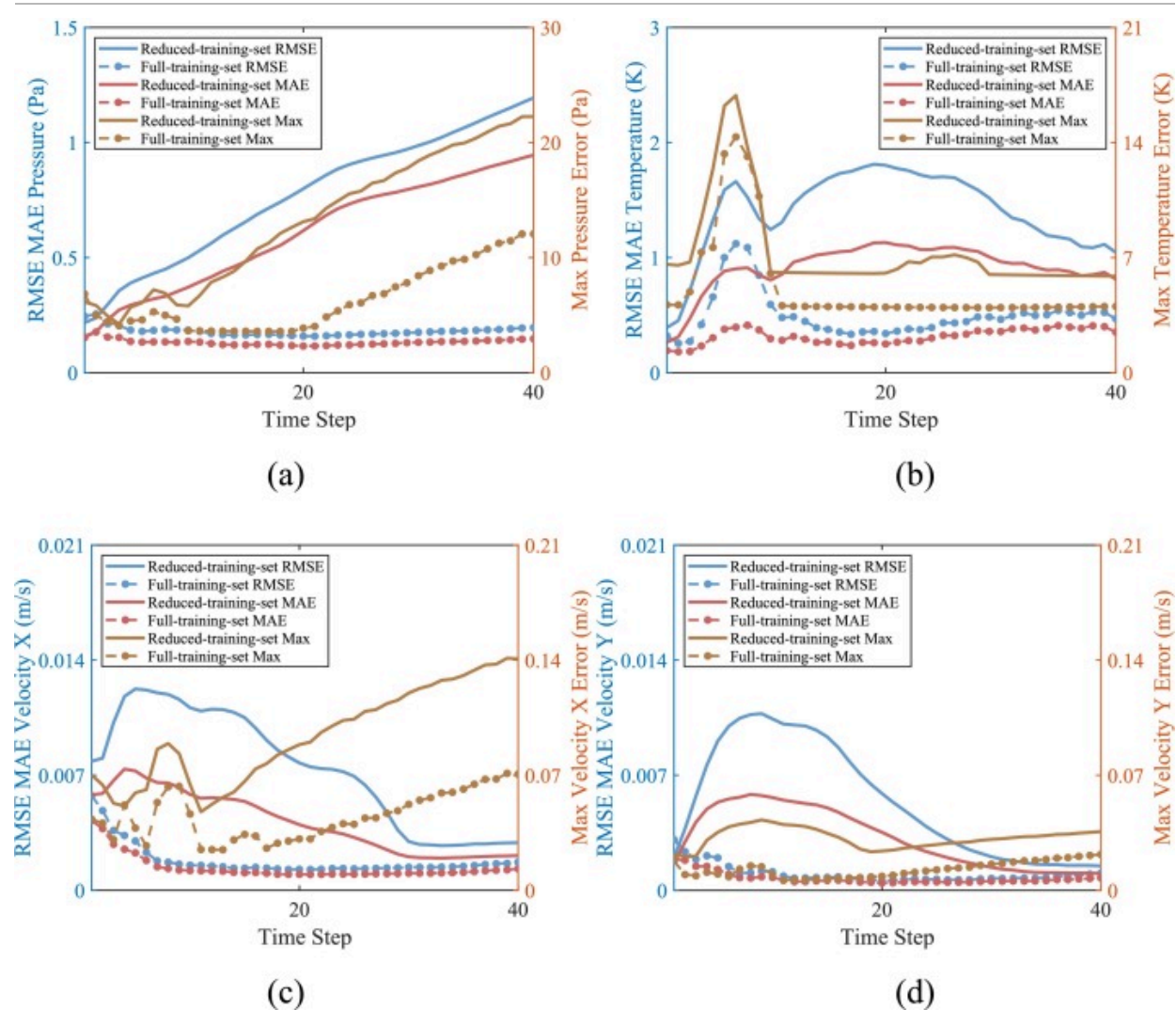
Fig. 11. RMSE and MAE with fixed weights vs. self-adaptive at varied resolutions: (a) Pressure error under fixed weights; (b) Pressure error under self-adaptive; (c) Temperature error under fixed weights; (d) Temperature error under self-adaptive; (e) Velocity X error under fixed weights; (f) Velocity X error under self-adaptive; (g) Velocity Y error under fixed weights; (h) Velocity Y error under self-adaptive.

Additional comparisons of the minimum and maximum errors under fixed and self-adaptive loss weights are provided in Supplementary Fig. S1. The results show that self-adaptive loss weights have only a minor effect on the minimum error, but substantially suppress the maximum error and error spikes, with the most pronounced improvement observed on the coarse 64×128 grid.

3.6. Influence of training set size

Under the same spatial resolution and time step settings, the influence of the training-set size is examined. Two training subsets containing 506 and 920 samples, respectively, are drawn from the full pool of 2760 samples, while the validation set is kept fixed. For clarity of presentation, the minimum errors are not plotted in Fig. 12 because their magnitudes

are very small. The results show that when the number of training samples increases from 506 to 920, the RMSE and MAE of the pressure field, temperature field, and velocity components all decrease systematically. The temporal fluctuation range of the errors is markedly reduced, and the maximum error consistently decreases as well. For the pressure and temperature fields, a larger dataset effectively suppresses the error accumulation over time observed under small-sample conditions. For the velocity fields, although the average errors are already relatively low, increasing the number of samples still leads to a substantial reduction in extreme errors. These results indicate that increasing the training data size improves prediction accuracy and reduces error propagation during autoregressive rollout.



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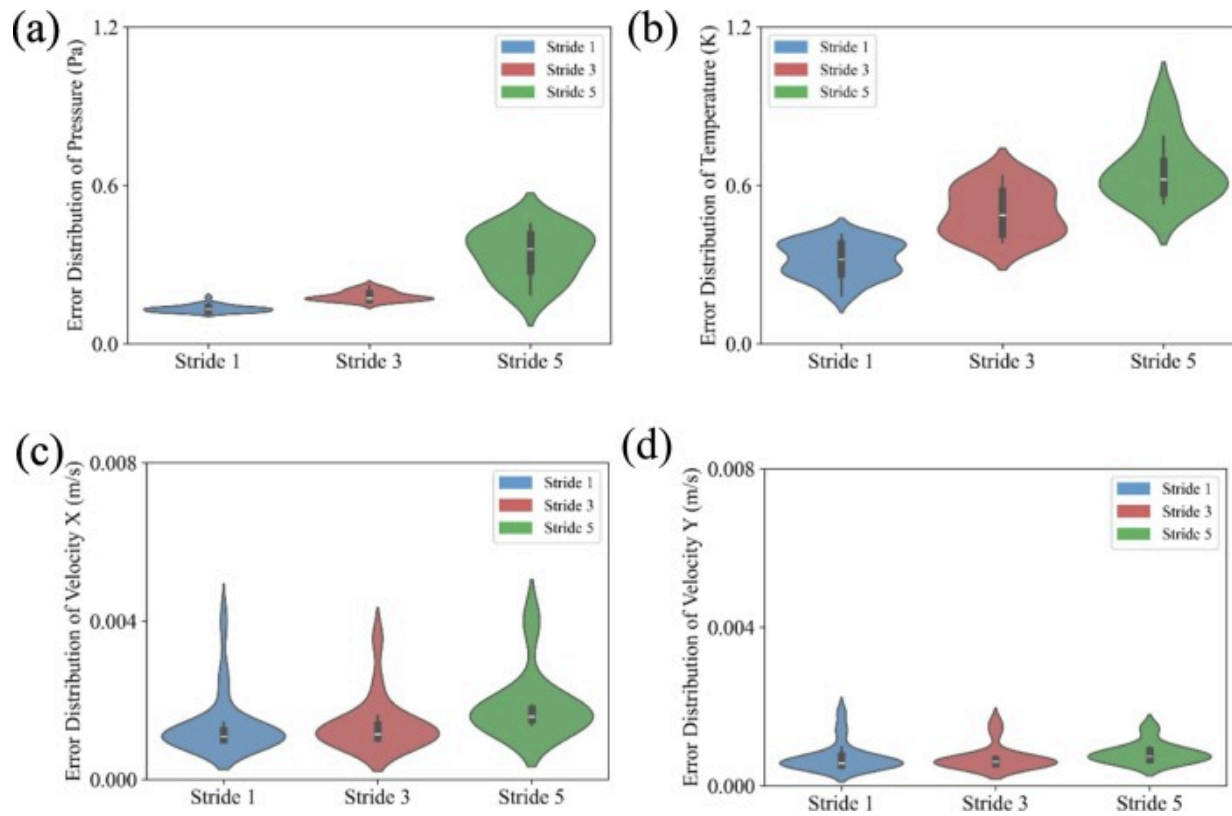
Fig. 12. Error comparison under reduced and full training dataset sizes: (a) Pressure; (b) Temperature; (c) Velocity X; (d) Velocity Y.

This section evaluates the computational efficiency of the proposed model under different prediction strides ($s = 1, 3, 5$) and compares it with FEM. The FEM solver advances the solution sequentially with a fixed time step Δt , whereas PAED-Net performs autoregressive rollout with stride s , so that one network prediction corresponds to the physical state separated by s FEM time steps. As a result, FEM efficiency is constrained by the admissible time step, while the inference time of PAED-Net remains nearly independent of stride. Table 4 summarizes the computation times at three spatial resolutions (64×128 , 128×256 , and 256×512). For FEM, the runtimes for one-, three-, and five-step advancement are 0.347s, 1.041s, and 1.735s, respectively, whereas PAED-Net requires only 0.006s, 0.0125s, and 0.042s for a single inference. All neural network inference times were measured on an NVIDIA GeForce RTX 3060 GPU using PyTorch with CUDA support, while the FEM simulations were executed on a CPU (Intel i9-10850K, 32GB RAM). Based on these results, PAED-Net achieves an acceleration of approximately 8–40 times over FEM for the Marangoni benchmark case. Moreover, when FEM suffers from data discontinuities or simulation breakdowns due to an inappropriate choice of time step, the proposed model still maintains stable inference performance. In practical deployment, PAED-Net requires only an initial physical field to start the autoregressive rollout, after which the subsequent temporal evolution is predicted entirely by the neural network.

Table 4. Comparison of computation time between FEM and PAED-Net.

Time	FEM (s)	64×128 (s)	128×256 (s)	256×512 (s)
Stride 1	0.347	0.006	0.0125	0.042
Stride 3	1.041	0.006	0.0127	0.043
Stride 5	1.735	0.0061	0.0128	0.043

Although data-driven models are not constrained by the strict time step stability conditions inherent to conventional FEM solvers, the choice of prediction stride still has an important impact on error evolution and predictive robustness. Fig. 13 shows the error distributions of the four physical fields for stride $s = 1, 3, 5$. For the pressure field, the error distribution becomes slightly more dispersed as the prediction stride increases, but the median error rises only marginally and the extreme values remain effectively controlled. The temperature field is the most sensitive to the stride: under Stride 1 and Stride 3, the error distribution is compact and the median remains low, whereas under Stride 5 the distribution broadens noticeably and the tail becomes significantly heavier. In contrast, the velocity fields are only weakly affected by the stride. The median error remains nearly unchanged, and under Stride 5 the distributions are only slightly wider. Overall, increasing the prediction stride has a limited effect on the median error of each physical field, while the control of extreme errors remains the primary concern. In the present task, Stride 3 offers a balanced compromise: it substantially improves throughput compared with Stride 1 while providing better control of tail risk in the temperature field than Stride 5.



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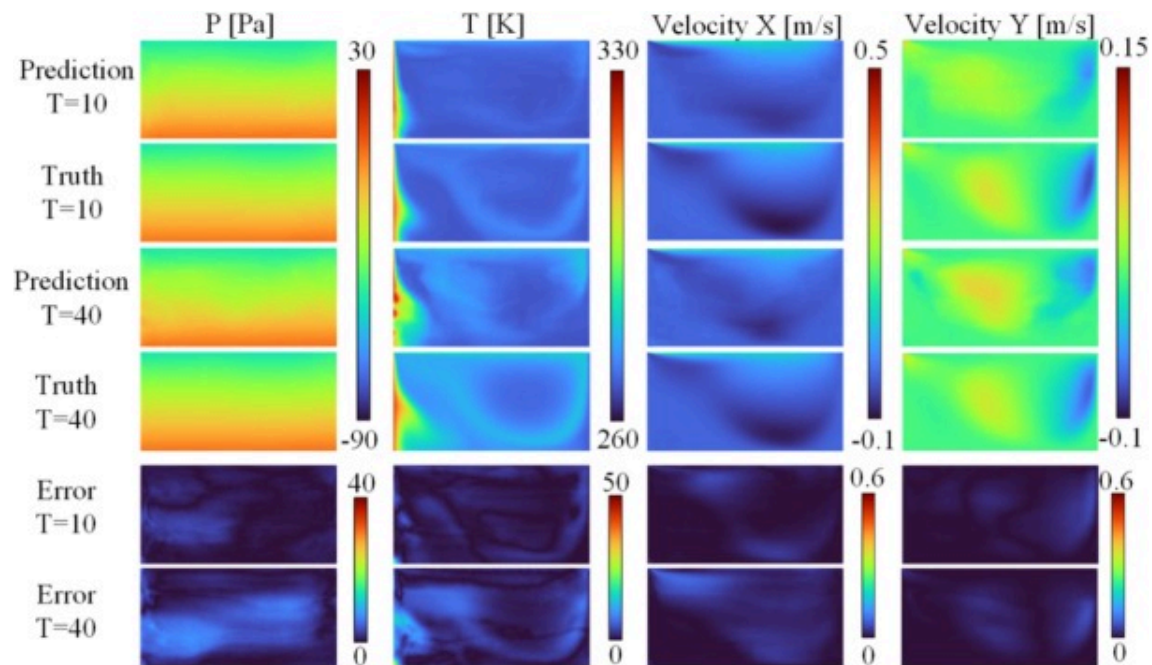
Fig. 13. Violin plots of error distributions for different strides: (a) Pressure; (b) Temperature; (c) Velocity X; (d) Velocity Y.

3.9. Generalization to different materials

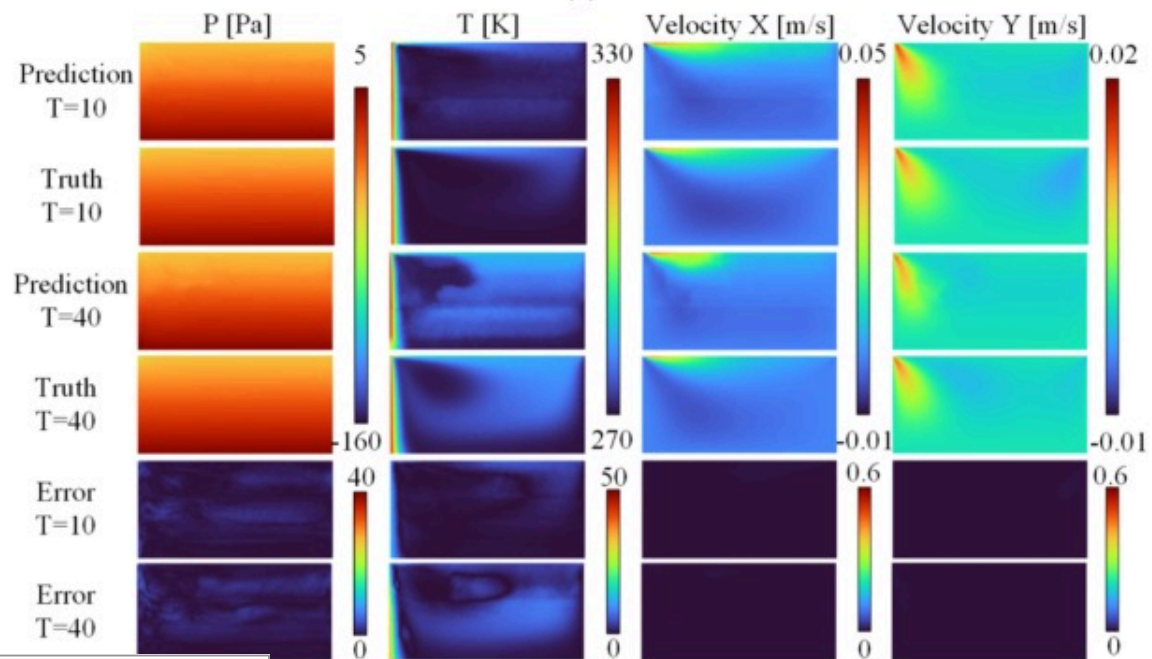
To assess the generalization capability of the proposed model across different fluid systems, PAED-Net is tested on Marangoni convection of water and ethylene glycol at a resolution of 256×512 , with a temperature difference of 50K imposed between the hot left wall and the cold right wall. Fig. 14 shows the predicted fields and corresponding four physical quantities at $t = 10$ and $t = 40$ for both fluids. Overall,

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the reconstructed fields of both water and ethylene glycol closely match the reference solution at $t = 10$. At $t = 40$, ethylene glycol still exhibits smooth error distributions with low amplitudes, whereas water develops stronger localized error patches and fine streaks around the temperature T and vertical velocity v_y , indicating that the high gradients and thin shear layers near the free surface in water are more prone to amplifying phase and amplitude deviations during long-horizon rollouts. This behavior can be attributed to two main factors. On the one hand, the higher viscosity of ethylene glycol damps the Marangoni-driven strong convection, leading to smoother flow patterns and more localized error fields. On the other hand, the lower viscosity of water enhances surface–bulk coupling, making error accumulation more likely in the later stages and allowing it to spread into the recirculation-closure region.



(a)



(b)

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Fig. 14. Predictions and errors for water and ethylene glycol at $t = 10$ and $t = 40$.

The detailed RMSE, MAE, and Max Error curves for water and ethylene glycol are provided in Supplementary Fig. S2, and the corresponding field ranges and relative errors are summarized in Tables S1–S2. These results show that ethylene glycol yields smoother and lower errors than water, while PAED-Net remains accurate across fluids with markedly different thermophysical properties.

4. Case study in laser powder bed fusion

To assess the applicability and predictive capability of the proposed PAED-Net in a representative laser-based additive manufacturing process, this section applies the model to a two-dimensional simulation of melt pool dynamics with a moving heat source in laser welding and additive manufacturing, building on our prior work [9], [49], [50], [51]. The laser welding and additive manufacturing process involves strongly coupled transient multiphysics, including heat conduction, melt flow, Marangoni convection, and solid–liquid phase change. In this case study, we focus on the specification of the heat source and boundary conditions; the governing equations and numerical treatment follow those used in the Marangoni benchmark. A prescribed inward heat flux is imposed on the top free surface. The total heat flux q_{total} consists of laser input, convective cooling, and radiative losses:

$$q_{\text{total}} = q_{\text{laser}}(x, y, t) + h_0 (T - T_{\infty}) + \varepsilon \sigma_{\text{SB}} (T^4 - T_{\infty}^4) \quad (26)$$

where $h_0 = 15 \text{ W}/(\text{m}^2 \cdot \text{K})$ is the convective heat transfer coefficient, T_{∞} is the ambient temperature, $\varepsilon = 0.3$ is the surface emissivity, and σ_{SB} is the Stefan–Boltzmann constant.

The laser heat source is modeled using a Gaussian distribution:

$$q_{\text{laser}}(x,y,t)=\eta(T)\frac{dP_l}{\pi R_0^2}\exp\left[-\frac{d}{R_0^2}\left((x-x_0-V_0t)^2+(y-y_0)^2\right)\right]$$

(27)

where P_l is the laser power, R_0 is the beam radius, d is the Gaussian shape factor, (x_0,y_0) denotes the initial laser position, and V_0 is the scan speed. The absorptivity $\eta(T)$ is modeled as a temperature-dependent function. In the present simulations, it increases with temperature and varies approximately within the range of 0.26–0.38.

Table 5 summarizes the key laser processing parameters and their ranges, including laser power, beam radius, Gaussian shape factor, and scan speed. These parameters jointly determine the spatial distribution and temporal evolution of the energy input and strongly influence the melt pool geometry and temperature field. The material considered here is QP980 steel, and its main thermophysical properties are listed in Table 6, including latent heat, solidus and liquidus temperatures, boiling point, surface tension and its temperature coefficient, thermal conductivity, and thermal diffusivity. In general, density, specific heat, and thermal conductivity are temperature dependent; however, in the present case they are assumed to be constant to simplify the model and facilitate a clearer comparison of the surrogate predictions.

Table 5. Heat source parameters.

Process parameters	Symbol	Value
Laser power	P_l	200W – 400W
Laser beam radius	R_0	0.4mm
Gaussian shape factor	d	2
Scan speed	V_0	2cm/s

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Material properties of QP980.

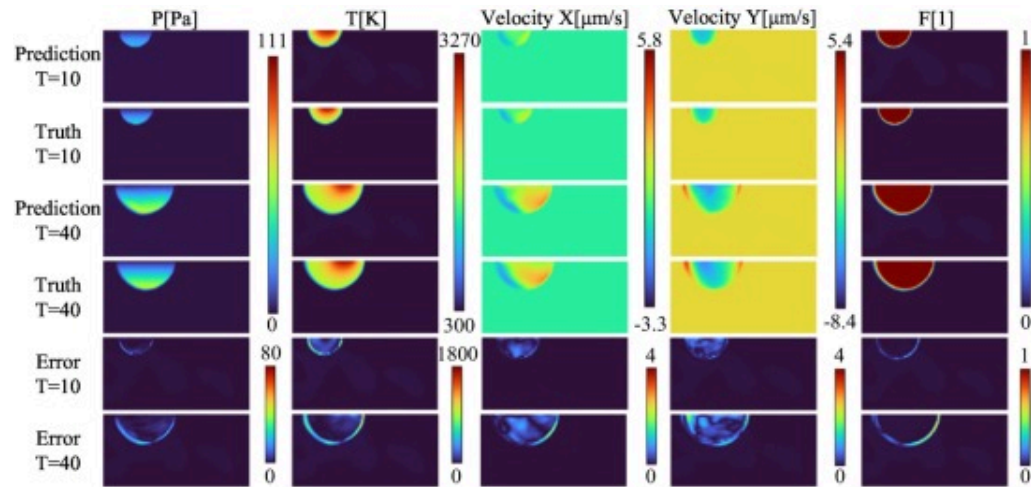
Properties	Symbol	QP980
Latent heat of fusion	L_f	250 kJ/kg
Latent heat of vaporization	ΔH_v	6.289 MJ/kg
Solidus temperature	T_s	1756 K
Liquidus temperature	T_m	1803 K
Boiling temperature	T_b	3273 K
Liquid surface tension	σ_m	1.61 N/m
Liquid surface tension gradient	$d\sigma/dT$	-4.47×10^{-4} N/m ² ·K
Thermal conductivity	k	43 W/m·K
Thermal diffusivity	D	11.72 mm ² /s
Molar gas constant	R	8.314 J/mol·K
Density	ρ	7.80 × 10 ³ kg/m ³
Specific heat	C_p	5.00 × 10 ² J/(kg·K)

To construct the training and test datasets, a two-dimensional computational domain of **5 mm × 2 mm** (length × thickness) is defined, and high-fidelity finite element simulations are performed on an unstructured mesh. In this study, the FEM simulations are used as reference data for training and evaluating the surrogate model. At the beginning of the simulation, the entire computational domain is initialized at the ambient temperature $T_0 = 300\text{ K}$, representing room-temperature conditions before laser irradiation. The mesh is locally refined along the laser scan path and in the expected melt pool region to resolve steep temperature gradients and flow structures. Each case is simulated for 600 ms with a time step of 10 ms, covering the complete thermal cycle of

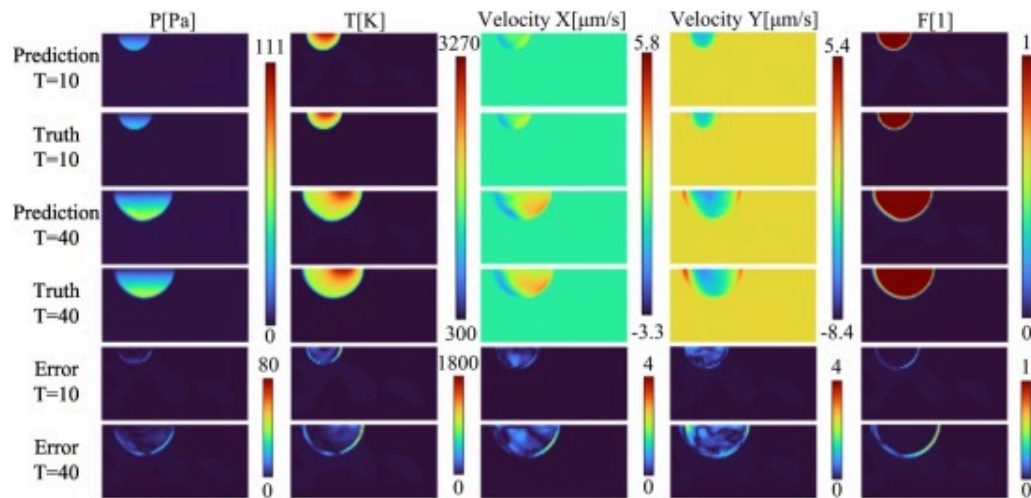
Typesetting math: 100% Compared with the Marangoni benchmark, an additional volume-fraction field is included. Accordingly, the inputs to PAED-Net consist of five physical fields at the

current time step (pressure, temperature, two velocity components, and volume fraction) together with a 10-dimensional parameter vector from the Physics-aware Subnet encoding laser power, scan speed, beam radius, material density, specific heat, thermal conductivity, surface-tension coefficient, temperature coefficient of surface tension, absorptivity, and molar gas constant. The model outputs the full-field distributions at the next time step. A total of 1100 samples are generated from the FEM simulations. The model is trained using one-step input–output pairs and evaluated through multi-step autoregressive rollout in temporal order. To further assess generalization, three intermediate laser powers (330W, 350W, and 370W) are excluded from the training and validation datasets and reserved exclusively for testing.

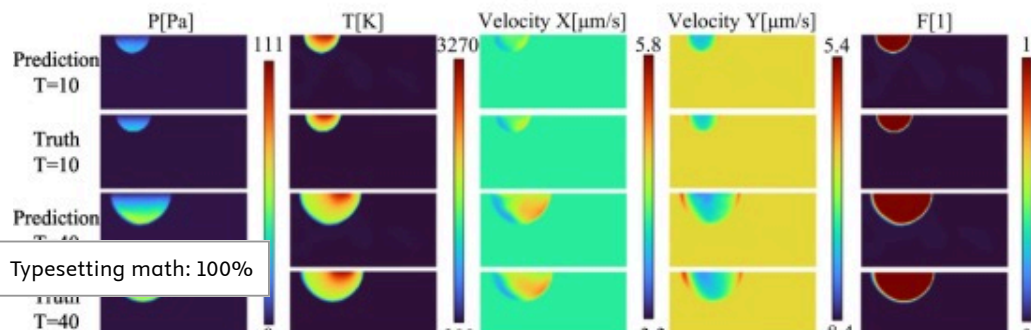
For three laser powers (330, 350, and 370W), the model successfully reproduces the characteristic spatio–temporal patterns of moving heat source, melt-pool response, and multiphysics coupling in laser welding and additive manufacturing, as shown in [Fig. 15](#). At time steps 10 and 40, the temperature peak moves along the scan path, and the front of the volume-fraction contour at $f = 0.5$ advances nearly synchronously. In the pressure field, a dome-shaped positive-pressure region forms upstream of the peak and moves with the laser trajectory, while a negative-pressure response is typically observed downstream. In the velocity field, the horizontal component forms a stable surface jet, whereas the vertical component exhibits a characteristic downwash–upwash pair near the hot spot, together forming a closed recirculation loop. As the laser power increases from 330W to 370W, the melt pool expands, the recirculation strengthens, the pressure dome sharpens, and the recirculation-closure region shifts slightly outward, indicating thinner thermal and shear boundary layers and stronger Marangoni driving. Apart from the melt-pool boundary, most of the domain maintains similar amplitudes and structures in pressure, temperature, and velocity, indicating that the surrogate captures the global flow topology well.



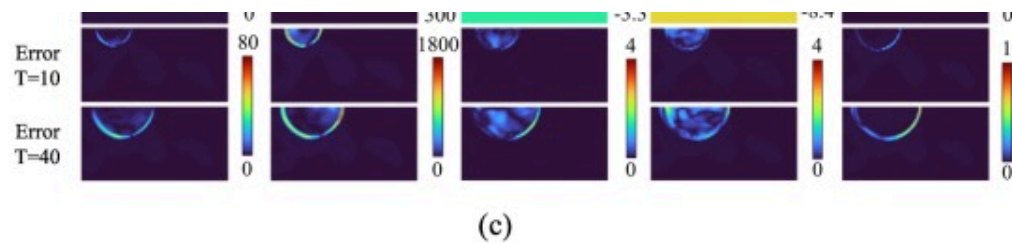
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(b)



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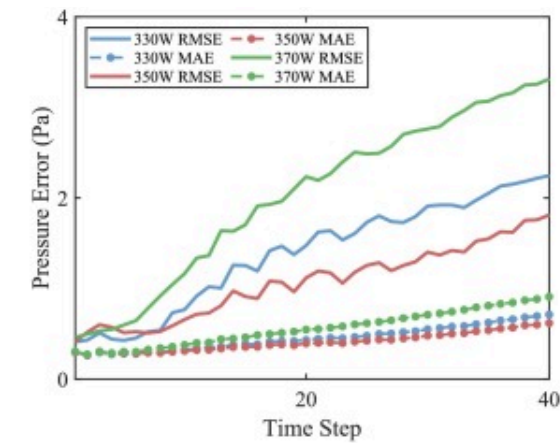
Fig. 15. Predictions and errors of QP980 at (a) 330W, (b) 350W, and (c) 370W.

The error distributions are qualitatively similar across the three powers and the two time instants: errors mainly form a continuous ring along the melt pool boundary and evolve from narrow/weak to wide/strong as the power and rollout step increase. Two main factors contribute to this behavior. Mechanistically, the largest temperature and velocity gradients are located at the free surface and solid–liquid interface, where the Marangoni shear layer and thermal boundary layer are thinnest. In these regions, even a spatial misalignment of one or two grid cells can produce a large numerical difference. This effect explains the bright error bands observed near the melt pool boundary, especially at step 40. Such deviations are most evident near the downwash/upwash regions and the recirculation-closure zone, where the solution is highly sensitive to interface shape and phase shifts. From the data-processing perspective, each physical field is multiplied by the volume fraction to emphasize liquid behavior. This operation introduces a sharp gradient at the solid–liquid interface. As a result, the prediction becomes more sensitive to small spatial deviations near the boundary. A comparison across fields shows that temperature and vertical velocity are the most sensitive, with high-intensity error patches appearing around the temperature peak and the downwash/upwash regions at time step 40. Pressure is less sensitive, with errors mainly concentrated near the leading-edge dome and sidewall corners, while the horizontal velocity is the most stable, indicating that the main jet topology is reliably captured. Overall, small phase lags in the

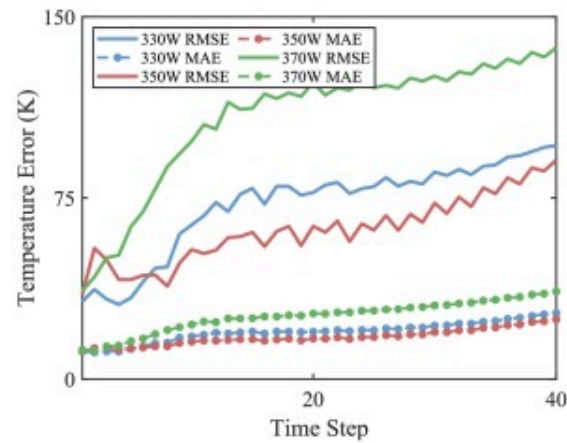
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recirculation-closure region and along key isotherms are the main source of late-stage error growth, whereas the overall amplitude level in the interior remains well preserved.

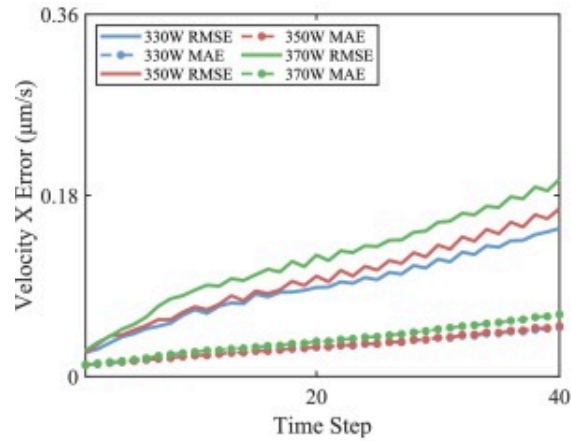
The error analysis reveals a clear dependence of the prediction accuracy on laser power, as well as a pronounced temporal accumulation effect. As shown in Fig. 16, for all physical fields, the 370W case exhibits the largest RMSE and MAE because it has the highest heat input and the thinnest thermal boundary layer. It is noteworthy that the overall errors at 330W are systematically higher than those at 350W, which can be attributed to three coupled mechanisms. First, when using the volume-fraction mask for evaluation, the melt pool region is defined using a threshold of $V_f > 0.5$. Under this criterion, the smaller melt pool at 330W leads to a larger statistical contribution of the boundary error band with a fixed thickness. Second, 350W is closer to the center of the training parameter space, where the model approximation is more robust. Third, for the smaller melt pool at 330W, the same pixel-scale phase lag in isotherm or interface position contributes more strongly to the RMSE. Across physical fields, the temperature field is the most sensitive to changes in power, followed by pressure and vertical velocity, whereas the horizontal velocity is predicted most robustly because of the relatively stable surface jet. For all fields, RMSE consistently exceeds MAE, indicating a heavy-tailed error distribution. The corresponding extreme errors are mainly located near the melt pool boundary, the leading-edge dome, and the downwash/upwash recirculation-closure region, where the solution is highly sensitive to interface position.



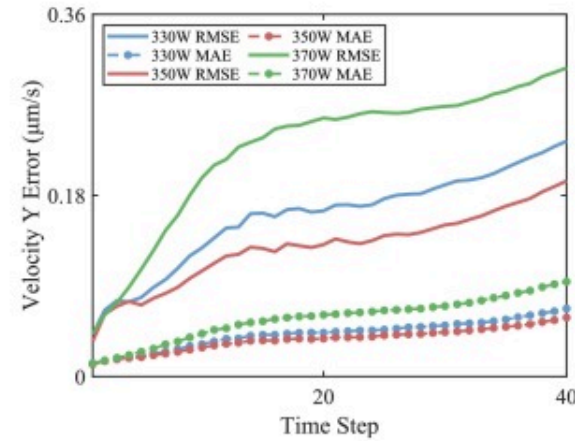
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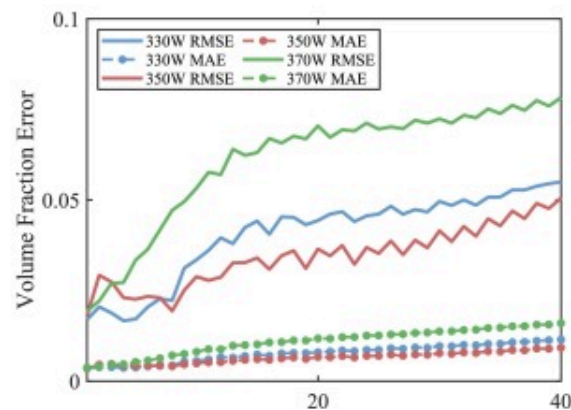
(b)



(c)



(d)



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Time Step
(e)

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Fig. 16. RMSE and MAE over 40 time steps of QP980: (a) Pressure; (b) Temperature; (c) Velocity X; (d) Velocity Y; (e) Volume fraction.

The detailed evolution of minimum and maximum errors under different laser powers is presented in Supplementary Fig. S3. Overall, the maximum error increases with laser power, whereas the minimum error remains close to zero. The temperature field is the most sensitive to power, while the horizontal velocity is the most stable, indicating that extreme-error control mainly depends on interface localization and phase consistency.

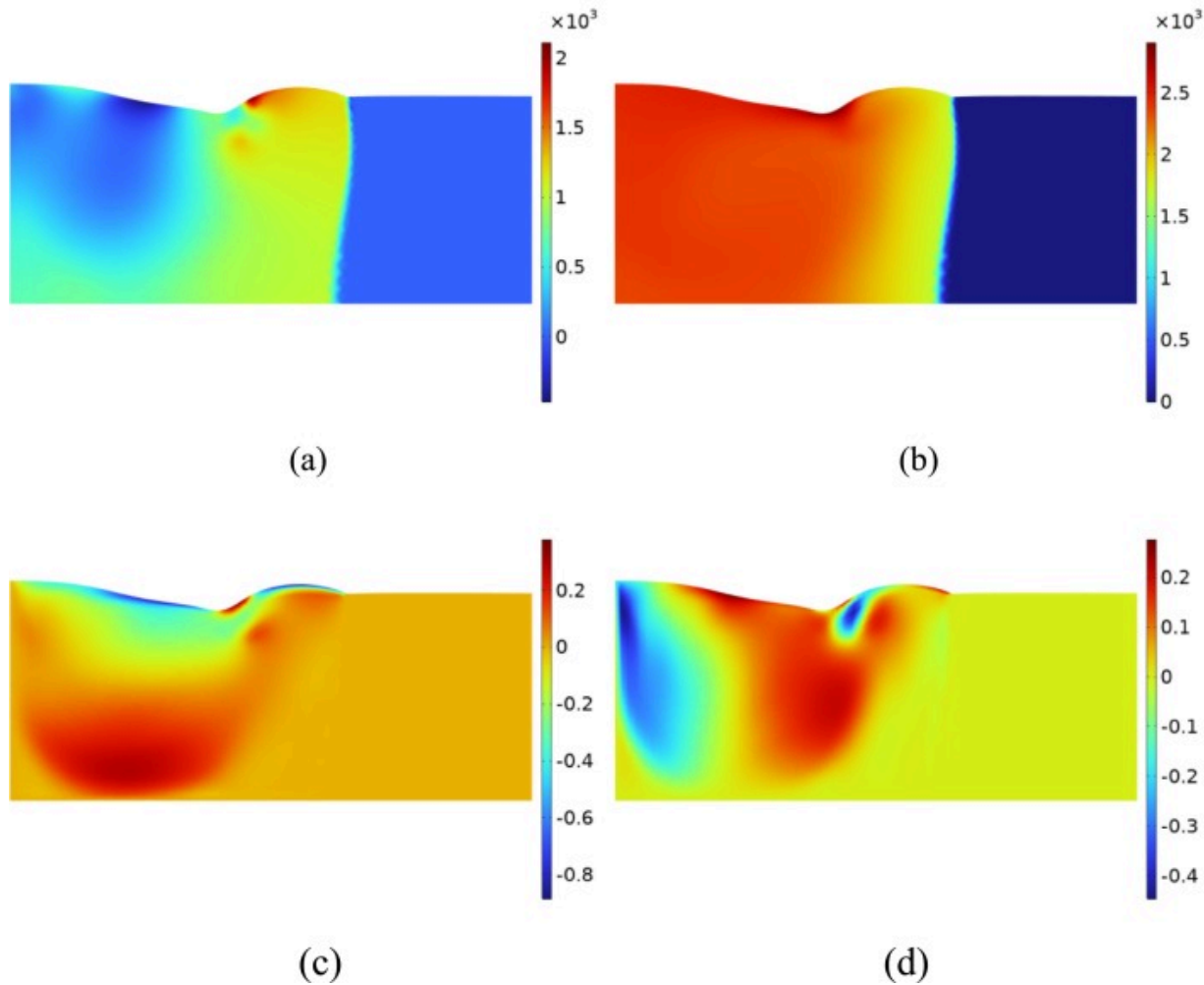
The RMSE over 40 time steps is averaged and normalized by the total range of each physical field to obtain a dimensionless mean relative error, as summarized in Table 7. Among the three power levels, the 370W case exhibits the largest errors, with mean relative errors of 3.3% and 6.1% for the temperature and volume-fraction fields, respectively, further indicating that higher power amplifies errors near interfaces and high-gradient regions. The 330W case ranks second, with corresponding errors of 2.2% and 4.1% for temperature and volume fraction. For the 350W case, all physical fields except v_x achieve the smallest errors; in particular, the mean errors of pressure and v_y are about 1%, consistent with the time-resolved error analysis. Overall, the volume fraction V_f and temperature field T are the most sensitive quantities, whereas the velocity and pressure fields are predicted more robustly. In general, PAED-Net demonstrates good robustness across different laser powers, with low overall errors and controlled error growth during long-horizon autoregressive prediction.

Table 7. Prediction errors of QP980.

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Physical fields	P	T	V _x	V _y	Vf
330W	1.3%	2.2%	1.1%	1.2%	4.1%
350W	1.0%	1.9%	1.2%	1.0%	3.4%
370W	1.9%	3.3%	1.5%	1.8%	6.1%

In this case, the FEM requires approximately 4min to advance a single 10ms time step, whereas PAED-Net takes only 0.006s, 0.0125s, and 0.042s per step at resolutions of **64 × 128**, **128 × 256**, and **256 × 512**, respectively. This corresponds to speed-up factors of about 40,000, 19,200, and 5714. Further tests show that when a free surface is explicitly taken into account, as illustrated in [Fig. 17](#), the FEM runtime per 10ms step increases to about 36min, while the inference time of PAED-Net remains essentially unchanged, leading to a maximum relative speed-up of up to about 360,000 times faster.



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Fig. 17. Numerical simulation of melt pool dynamics with a free surface.

5. Conclusions

In this study, a DAED-Net is proposed for efficient and accurate prediction of multiphysics welding and additive manufacturing processes. Based on systematic

case studies of Marangoni convection and laser welding and additive manufacturing, the following conclusions can be drawn:

- (1) Under a wide range of operating conditions and material systems, PAED-Net achieves relative errors mostly within 5%–8% for pressure, temperature, and velocity fields, while accurately capturing key physical features such as melt pool geometry, temperature gradients, and surface recirculation patterns. The self-adaptive weighted loss effectively balances the different magnitudes of multiple physical fields, and the autoregressive training scheme significantly improves temporal stability and enhances the representation of high-gradient regions.
- (2) For the Marangoni benchmark, PAED-Net delivers a speed-up of approximately 8–40 times compared with a conventional FEM solver. In the laser welding and additive manufacturing case, the acceleration reaches about 40,000, 19,200, and 5714 times at spatial resolutions of 64×128 , 128×256 , and 256×512 , respectively. When free-surface evolution is explicitly taken into account, the speed-up can be further increased to around 360,000 times, while the inference time of PAED-Net remains nearly unchanged. The inference cost is essentially independent of the prediction horizon, indicating strong potential for real-time applications such as digital twin systems.
- (3) Test results on silicone oil, water, ethylene glycol, and QP980 steel demonstrate that PAED-Net maintains stable predictive performance under diverse combinations of thermophysical properties and process parameters. This confirms the effectiveness of the Physics-aware Subnet in embedding prior physics and provides a solid basis for extending the model to a broader range of laser processing scenarios.

Overall, PAED-Net provides a data-driven surrogate that achieves a favorable balance between accuracy and efficiency for laser processing simulations and, compared with full-field PDE solvers, relies on a single offline training stage and end-to-end mapping, thereby simplifying the multiphysics modeling workflow.

Despite the advantages demonstrated in this study, several limitations remain:

- (1) Similar to many autoregressive surrogate models for time-dependent physical systems, prediction errors may accumulate during long-horizon rollout because the predicted state at each step is recursively used as the input for the next step. Although the model remains stable within the prediction horizons considered in this study, reliability is mainly ensured within a finite time window. Possible mitigation strategies include scheduled sampling during training, periodic correction using numerical simulations or sensor observations, and the integration of data assimilation techniques.
- (2) The current implementation requires an initial physical state to start the autoregressive rollout. In this study, the starting field is obtained from the first simulated time step generated by the FEM solver, since directly starting from the initial condition leads to larger error accumulation. In practical digital-twin scenarios, such initial states could also be obtained from experimental measurements, sensor data, or state-estimation techniques.
- (3) In the present simulations, density, specific heat, and thermal conductivity are assumed to be constant for simplicity. In practical LPBF and welding processes, these properties may vary significantly with temperature. Nevertheless, the proposed PAED-Net framework is not restricted to constant-property datasets, since temperature-dependent properties are commonly represented as functions of temperature in FEM simulations and learned by the neural network from the training data.

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Future work will focus on automatically identifying multiphysics coupling structures from governing equations, incorporating real-time sensing data with online correction and data assimilation, and extending the present two-dimensional framework to fully three-dimensional multiphysics simulations.

CRediT authorship contribution statement

Zhehao Zhang: Writing – original draft, Methodology, Investigation. **Zihan Wang:** Visualization, Formal analysis. **Xuefei Cui:** Visualization, Conceptualization. **Yi Zheng:** Writing – review & editing, Conceptualization. **Daqing Wang:** Supervision, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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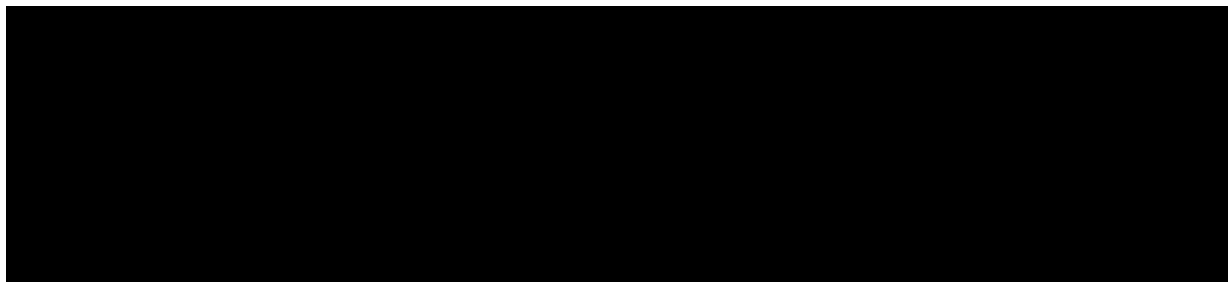
Appendix A. Supplementary data

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Supplementary material



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Supplementary Video S1. Marangoni convection forecast video

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